

TCS 3000 REMOTE INTERFACE Protocol V1.4.4.1002 / 10.02.2019



CONFIDENTIAL

REVISION HISTORY

REV No.	DATE	DESCRIPTION	Author(s)
0.1...0.7		V0 - >V1 protocol definition and updates	EBM
1.3.594	27.07.2012	Updated to protocol version 1, synchronized with TCS3000 V1.3.1.594	EBM
1.3.599	30.07.2012	Added note related to byte order in the data	EBM
1.3.602	01.08.2012	Corrected some descriptions, Switched to Set/Get (0x20/0x40) ONLY flag system, Added commands 0x0E,0x0F,0x10,0x11,0x35	EBM
1.3.608	02.08.2012	Updated ERRORCODES, Added Algorithm Examples	EBM
1.3.617	06.08.2012	Added commands 0x43 related to NoFlow Timer, Updated Errorcodes (added ERR 0x0F)	EBM
1.3.659	06.09.2012	Fixed valid addresses range, Added errorcodes for interface busy and data integrity error, added connection examples, added interface timing description and examples, added description of the STATUS byte	EBM
1.3.662	10.09.2012	Updated interface marking in the connection diagrams to match the notation used on the TCS3000 Device, Minor corrections in the text	EBM
1.4.0.687	06.01.2013	Added Status "Multiple Pending" for multiple deliveries	EBM
1.4.4.1001	10.02.2019	Updated command list to the most current one, updated errorcodes	EBM

CONVENTIONS

Command Fields	
<DATA>	COMMAND: Data Field in a command (length N bytes)
BTCNT	COMMAND: Data Package length
CMD	COMMAND: Command Code
CRC	COMMAND: Checksum (ALWAYS using DOW CRC8)
DEST	COMMAND: Destination Address
ERR	COMMAND: Errorcode
FLAG	COMMAND: Flag byte (From HOST)
HEADER	COMMAND: Header (in the Remote interface it is always 0x7E)
RFLAG	COMMAND: RETURN Flags (From TCS)
SRC	COMMAND: Source Address
STAT	COMMAND: System Status byte - currently unused (always 0)

Datatypes	
U8	Unsigned 8 bit Integer (Unsigned Byte)
I8	Signed 8 bit Integer (Byte)
U16	Unsigned 16 bit Integer (Unsigned Word)
I16	Signed 16 bit Integer (Word)
U32	Unsigned 32 bit integer (Unsigned Long)
I32	Signed 32 bit Integer (Long)
U64	Unsigned 64 bit integer (Unsigned Quad)
I64	Signed 64 bit Integer (Quad)
Dbl	Double Precision floating point (64bit IEEE754)
ENUM	Enumerated variable (always U8, 0 means ERROR) see ENUMS.ini for details
String	String (either normal ASCII string or combination of more than one datatype)

IMPORTANT NOTE:

Difference in NUMERIC DATA FORMATS used by Intel x86 Processors and TCS3000

Integers (U16/I16; U32 / I32; U64 / I64), and Floats (Double Precision – 64 bits) are stored in TCS3000 with their Most Significant Byte (MSB) at the lowest address (LSB -being stored in the highest address). This is the opposite of the way Intel x86 stores its numeric data.

Example: The number “180” (unsigned 16 Bit Integer Value) is stored the following way:

- In TCS 3000: “**0x00B4**” In the same sequence, this data will be transmitted from TCS3000 through RS232 interface to external Computer (00h followed by B4h.)
- In INTEL x86 /Windows/ Machine : “**0xB400**”
To transmit this data correctly to TCS3000, the Byte Order must be reversed.

COMMAND - HOST to TCS3000 (DOWNLINK)

FIELDS DESCRIPTION

Sender / Receiver addressing:	
Address	Device
00	HOST
01	Default TCS3000 Address
01...127	Valid slave addresses
250...255	RESERVED

FIELDS DESCRIPTION

Byte 0	HEADER								Transfer Header - Always 0x7E (01111110b)	
Byte 1		DEST							Receiver address -> HOST=00, TCS=01, 02...slave Nr	
Byte 2			SRC						Sender address -> HOST=00, TCS=01, 02...slave Nr	
Byte 3				RFLAG					Command Flags 8-bits (see description)	
Byte 4					CMD				Command Code - Same as command from HOST	
Byte 5						BTCNT			Data package length in bytes (N+2=0...249)	
Byte 6							ERR		TCS3000 RI ERROR CODE (0 = NO ERROR)	
Byte 7								STAT	STATUS	
Byte (N+8)									Checksum (DOW CRC8)	CRC

FLAGS DESCRIPTION

COMMAND - HOST to TCS3000 (DOWNLINK)

COMMAND - 6 Bytes						CMD DATA	CRC
Byte 0	Byte 1	Byte 2	Byte 3	Byte 4	Byte 5	N Bytes	Byte (N+6)
0X7E	DEST	SRC	FLAG	CMD	BTCNT	<DATA>	CRC

COMMAND FLAG BITS DESCRIPTION				
Byte	Bit	HEX	FLAG	Description
3	0	0x01	N/A	Don't Care
3	1	0x02	N/A	Don't Care
3	2	0x04	N/A	Don't Care
3	3	0x08	N/A	Don't Care
3	4	0x10	N/A	Don't Care
3	5	0x20	SET	SET (or SET COMMAND)
3	6	0x40	GET	GET (or GET COMMAND)
3	7	0x80	N/A	Don't Care

COMMAND ANSWER - TCS3000 to HOST (UPLINK)

COMMAND - 8 Bytes								CMD DATA	CRC
Byte 0	Byte 1	Byte 2	Byte 3	Byte 4	Byte 5	Byte 6	Byte 7	N Bytes	Byte (N+8)
0X7E	DEST	SRC	RFLAG	CMD	BTCNT	ERR	STAT	<DATA>	CRC

ANSWER FLAG BITS DESCRIPTION				
Byte	Bit	HEX	FLAG	Description
3	0	0x01	CREC	CMD Recognized
3	1	0x02	OPOK	Operation OK
3	2	0x04	OK	ALL OK
3	3	0x08	N/A	SAME AS ORIG. CMD
3	4	0x10	N/A	SAME AS ORIG. CMD
3	5	0x20	SET	SAME AS ORIG. CMD
3	6	0x40	GET	SAME AS ORIG. CMD
3	7	0x80	ACK	Transfer Acknowledge

In case of CRC Error or incorrect package the 0x80 flag = 0

STATUS BYTE DESCRIPTION			
Byte	Bits	Value	Description
7	<7:4> DELIVERY STATUS	0	ERROR
		1	DELIVERY IDLE
		2	DELIVERY ACTIVE
		3	AIR DETECTED
		4	DELIVERY PAUSED
		5	DELIVERY STOPPED
		6	TICKET PENDING
		7	PRINTING
7	<3:0> SYSTEM STATUS	8	MULTIPLE PENDING
		0	ERROR
		1	SYSTEM IDLE
		2	W&M
		3	DELIVERY ACTIVE
		4	SYSTEM BUSY

<- Added in version 687

CRC Computation Details

The CRC of a command is calculated over the whole package (except the CRC itself which is the last byte in the transfer)
As shown in the examples below the CRC is calculated over the whole "green / shaded" area

COMMAND - HOST to TCS3000 (DOWNLINK)							
0X7E	DEST	SRC	FLAG	CMD	BTCNT	<DATA>	CRC

COMMAND ANSWER - TCS3000 to HOST (UPLINK)							
0X7E	DEST	SRC	FLAG	CMD	BTCNT	ERR	STAT
<DATA>							CRC

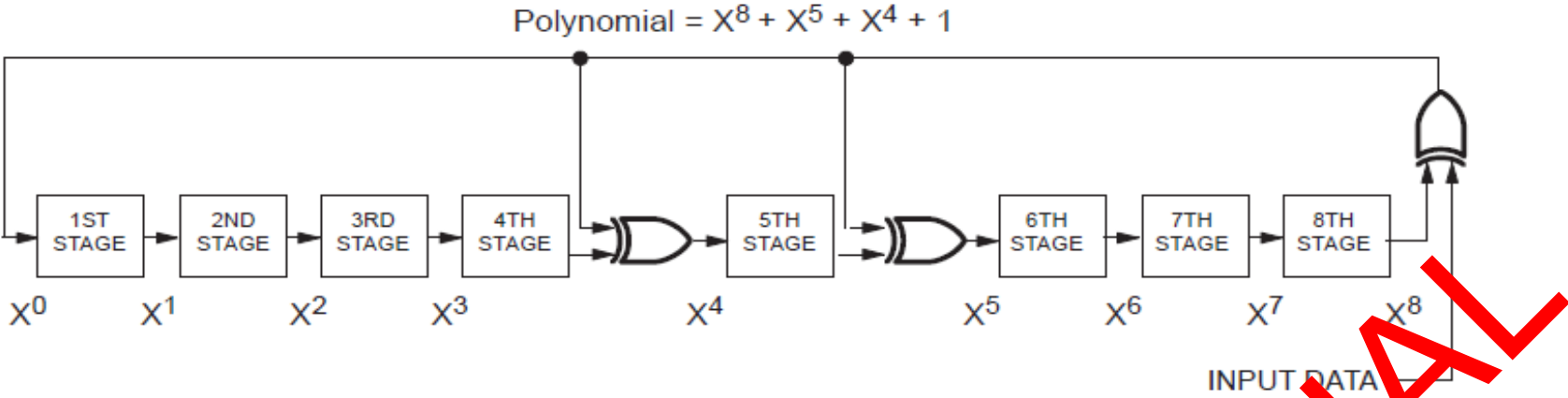
When receiving an Answer to a command (from the Uplink) if the Host computes the CRC over the whole transfer including the CRC byte the result of this computation should be 0 (zero) indicating correct transfer

The CRC Algorithm used is DOW CRC 8 (Dallas Semiconductor 1-Wire CRC 8) which is somewhat more complex than the "standard" CRC Algorithms and provides slightly better single bit error detection.

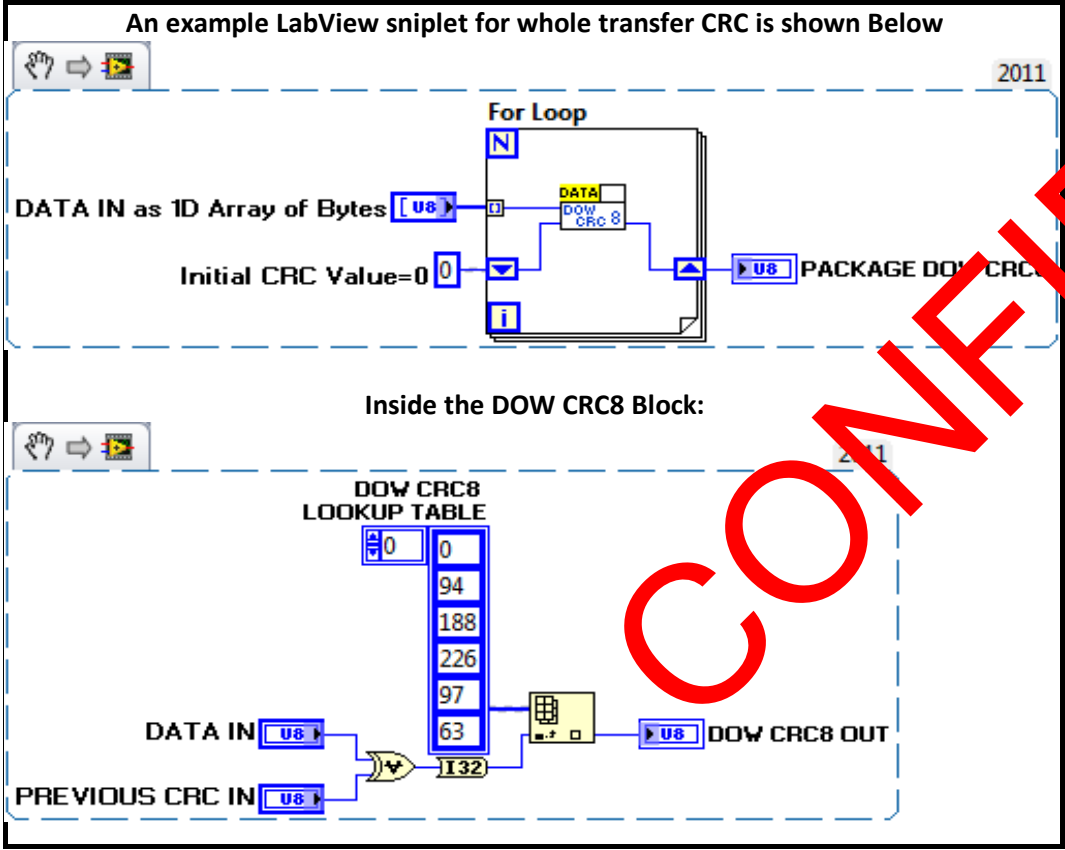
Short Description of DOW CRC 8 (for very detailed description refer to document AN937 pp133-139):

Polynomial Representation: $CRC\ Poly = X^8 + X^5 + X^4 + 1$

Algorithmic Representation:



However due to it's relatively high complexity and computation intensity the best approach is to replace this algorithm with Lookup table and XOR the next byte with the last calculated CRC and use the result as an index.



C Code generating lookup table and using it for CRC generation/calculation:

```
const unsigned char crc_array[]=
{
  0, 94, 188, 226, 97, 63, 221, 131, 194, 156, 126, 32, 163, 253, 31, 65,
  157, 195, 33, 127, 252, 162, 64, 30, 95, 1, 227, 189, 62, 96, 130, 220,
  35, 125, 159, 193, 66, 28, 254, 160, 225, 191, 93, 3, 128, 222, 60, 98,
  190, 224, 2, 92, 223, 129, 99, 61, 124, 34, 192, 158, 29, 67, 161, 255,
  70, 24, 250, 164, 39, 121, 155, 197, 132, 218, 56, 102, 229, 187, 89, 7,
  219, 133, 103, 57, 186, 228, 6, 88, 25, 71, 165, 251, 120, 38, 196, 154,
  101, 59, 217, 135, 4, 90, 184, 230, 167, 249, 27, 69, 198, 152, 122, 36,
  248, 166, 68, 26, 153, 199, 37, 123, 58, 100, 134, 216, 91, 5, 231, 185,
  140, 210, 48, 110, 237, 179, 81, 15, 78, 16, 242, 172, 47, 113, 147, 205,
  17, 79, 173, 243, 112, 46, 204, 146, 211, 141, 111, 49, 178, 236, 14, 80,
  175, 241, 19, 77, 206, 144, 114, 44, 109, 51, 209, 143, 12, 82, 176, 238,
  50, 108, 142, 208, 83, 13, 239, 177, 240, 174, 76, 18, 145, 207, 45, 115,
  202, 148, 118, 40, 171, 245, 23, 73, 8, 86, 180, 234, 105, 55, 213, 139,
  87, 9, 235, 181, 54, 104, 138, 212, 149, 203, 41, 119, 244, 170, 72, 22,
  233, 183, 85, 11, 136, 214, 52, 106, 43, 117, 151, 201, 74, 20, 246, 168,
  116, 42, 200, 150, 21, 75, 169, 247, 182, 232, 10, 84, 215, 137, 107, 53
}; // crc_table[]

unsigned char calc_checksum_8(unsigned char * data, unsigned int len)
{
  unsigned char crc = 0;
  for(unsigned int i=0; i < len; i++)
  {
    crc = crc_array[ crc ^ data[i] ];
  }
  return crc;
}
```

CONFIDENTIAL

TCS 3000 REMOTE INTERFACE Protocol V1.4.4.1002 / 10.02.2019

DOW CRC8 LOOKUP TABLE

Index	Value	Index	Value	Index	Value	Index	Value	Index	Value	Index	Value	Index	Value	Index	Value
0	0	32	35	64	70	96	101	128	140	160	175	192	202	224	233
1	94	33	125	65	24	97	59	129	210	161	241	193	148	225	183
2	188	34	159	66	250	98	217	130	48	162	19	194	118	226	85
3	226	35	193	67	164	99	135	131	110	163	77	195	40	227	11
4	97	36	66	68	39	100	4	132	237	164	206	196	171	228	136
5	63	37	28	69	121	101	90	133	179	165	144	197	245	229	214
6	221	38	254	70	155	102	184	134	81	166	114	198	23	230	52
7	131	39	160	71	197	103	230	135	15	167	44	199	73	231	106
8	194	40	225	72	132	104	167	136	78	168	109	200	8	232	43
9	156	41	191	73	218	105	249	137	16	169	51	201	86	233	117
10	126	42	93	74	56	106	27	138	242	170	209	202	180	234	151
11	32	43	3	75	102	107	69	139	172	171	143	203	234	235	201
12	163	44	128	76	229	108	198	140	47	172	12	204	105	236	74
13	253	45	222	77	187	109	152	141	113	173	82	205	55	237	20
14	31	46	60	78	89	110	122	142	147	174	176	206	213	238	246
15	65	47	98	79	7	111	36	143	205	175	238	207	139	239	168
16	157	48	190	80	219	112	248	144	17	176	50	208	87	240	116
17	195	49	224	81	133	113	166	145	79	177	108	209	9	241	42
18	33	50	2	82	103	114	68	146	173	178	142	210	235	242	200
19	127	51	92	83	57	115	26	147	243	179	208	211	181	243	150
20	252	52	223	84	186	116	153	148	112	180	83	212	54	244	21
21	162	53	129	85	228	117	199	149	46	181	13	213	104	245	75
22	64	54	99	86	6	118	37	150	204	182	239	214	138	246	169
23	30	55	61	87	88	119	123	151	146	183	177	215	212	247	247
24	95	56	124	88	25	120	58	152	211	184	240	216	149	248	182
25	1	57	34	89	71	121	100	153	141	185	174	217	203	249	232
26	227	58	192	90	165	122	134	154	111	186	76	218	41	250	10
27	189	59	158	91	251	123	216	155	49	187	18	219	119	251	84
28	62	60	29	92	120	124	91	156	178	188	145	220	244	252	215
29	96	61	67	93	38	125	5	157	236	189	207	221	170	253	137
30	130	62	161	94	196	126	231	158	14	190	4	222	72	254	107
31	220	63	255	95	154	127	185	159	80	191	111	223	22	255	53

Command Code		Command Flags	
0x02		0x20	
Code (dec): 2		Flags (dec): 32	

Save System State

16bit Identifier: 0x0220

Forces the TCS3000 to save it's current state

Modes: ALL ✓ IDLE ✓ W&M ✓ DLV ✓ BUSY ✓

COMMAND: HOST -> TCS3000 (Downlink)

Datatype:	N/A
Datalength:	0 Bytes
Host data description: No data required from host	

ANSWER: TCS3000 -> HOST (Uplink)

Datatype:	ACKNOWLEDGE
Datalength:	0 Bytes
Return data description: Command returns only ERRORCODE and STATUS	

CONFIDENTIAL

Command Code		Command Flags	
0x03		0x40	
Code (dec): 3		Flags (dec): 64	

GET System Mode

16bit Identifier: 0x0340

Asks the current system state (mode) from the TCS3000

Modes: ALL ✓ IDLE ✓ W&M ✓ DLV ✓ BUSY ✓

COMMAND: HOST -> TCS3000 (Downlink)

Datatype:	N/A
Datalength:	0 Bytes
Host data description: No data required from host	

ANSWER: TCS3000 -> HOST (Uplink)

Datatype:	ENUM
Datalength:	1 Bytes
Return data description: Returns the system state as ENUM(SYSSTATE): 0=ERROR 1=IDLE 2=WM 3=DLV_ACTIVE 4=BUSY	

CONFIDENTIAL

Command Code

0x04

Code (dec): 4

Command Flags

0x20

Flags (dec): 32

SET Time

16bit Identifier: 0x0420

Sets the system time of TCS3000

Modes:

ALL ✖

IDLE ✔

W&M ✔

DLV ✖

BUSY ✖

COMMAND: HOST -> TCS3000 (Downlink)

Datatype:

String

Datalength:

6 Bytes

Host data description: The system time as STRING in format: HHMMSS

ANSWER: TCS3000 -> HOST (Uplink)

Datatype:

ACKNOWLEDGE

Datalength:

0 Bytes

Return data description: Command returns only ERRORCODE and STATUS

Command Code

0x04

Code (dec): 4

Command Flags

0x40

Flags (dec): 64

GET Time

16bit Identifier: 0x0440

Asks the system time from TCS3000

Modes:

ALL ✔

IDLE ✔

W&M ✔

DLV ✔

BUSY ✔

COMMAND: HOST -> TCS3000 (Downlink)

Datatype:

N/A

Datalength:

0 Bytes

Host data description: No data required from host

ANSWER: TCS3000 -> HOST (Uplink)

Datatype:

String

Datalength:

6 Bytes

Return data description: The system time as STRING in format: HHMMSS

CONFIDENTIAL

Command Code

0x05

Code (dec): 5

Command Flags

0x20

Flags (dec): 32

SET Date

16bit Identifier: 0x0520

Sets the system date of TCS3000

Modes: ALL ✗ IDLE ✓ W&M ✓ DLV ✗ BUSY ✗

COMMAND: HOST -> TCS3000 (Downlink)

Datatype:

String

Datalength:

8 Bytes

Host data description: The system date as STRING in format: YYYYMMDD

ANSWER: TCS3000 -> HOST (Uplink)

Datatype:

ACKNOWLEDGE

Datalength:

0 Bytes

Return data description: Command returns only ERRORCODE and STATUS

Command Code

0x05

Code (dec): 5

Command Flags

0x40

Flags (dec): 64

GET Date

16bit Identifier: 0x0540

Asks the system date from TCS3000

Modes: ALL ✓ IDLE ✓ W&M ✓ DLV ✓ BUSY ✓

COMMAND: HOST -> TCS3000 (Downlink)

Datatype:

N/A

Datalength:

0 Bytes

Host data description: No data required from host

ANSWER: TCS3000 -> HOST (Uplink)

Datatype:

String

Datalength:

8 Bytes

Return data description: The system date as STRING in format: YYYYMMDD

CONFIDENTIAL

Command Code

0x06

Code (dec): 6

Command Flags

0x40

Flags (dec): 64

Get System Metrics

16bit Identifier: 0x0640

Returns the TCS3000 System Totalizers Volume unit

Modes: ALL ✓ IDLE ✓ W&M ✓ DLV ✓ BUSY ✓

COMMAND: HOST -> TCS3000 (Downlink)

Datatype:

N/A

Datalength:

0 Bytes

Host data description:

ANSWER: TCS3000 -> HOST (Uplink)

Datatype:

ENUM

Datalength:

1 Bytes

Return data description: The System Totalizers Unit as ENUM(RI_VOLMETRICS)

0=ERROR

1=GAL

2=L

3=UKG

4=daL

5=dL

6=cL

7=mL

8=m3

9=cm3

10=bbl

11=floz

12=ft3

13=in3

14=NOUNIT

CONFIDENTIAL

Command Code	Command Flags	
0x07	0x20	Set System Precision
Code (dec): 7	Flags (dec): 32	16bit Identifier: 0x0720

Sets the display/ticket precision of TCS3000

Modes:	ALL ✖	IDLE ✖	W&M ✔	DLV ✖	BUSY ✖
--------	-------	--------	-------	-------	--------

COMMAND: HOST -> TCS3000 (Downlink)

Datatype:	ENUM
Datalength:	1 Bytes
Host data description: The system precision as ENUM(RI_SYSDEC) 0=ERROR 1=1 2=1.1 3=1.11 4=1.111	

ANSWER: TCS3000 -> HOST (Uplink)

Datatype:	ACKNOWLEDGE
Datalength:	0 Bytes
Return data description: Command returns only ERRORCODE and STATUS	

Command Code	Command Flags	
0x07	0x40	Get System Precision
Code (dec): 7	Flags (dec): 64	16bit Identifier: 0x0740

Asks the display/ticket precision from TCS3000

Modes:	ALL ✔	IDLE ✔	W&M ✔	DLV ✔	BUSY ✔
--------	-------	--------	-------	-------	--------

COMMAND: HOST -> TCS3000 (Downlink)

Datatype:	N/A
Datalength:	0 Bytes
Host data description: No data required from host	

ANSWER: TCS3000 -> HOST (Uplink)

Datatype:	String
Datalength:	6 Bytes
Return data description: The system precision as ENUM(RI_SYSDEC) 0=ERROR 1=1 2=1.1 3=1.11 4=1.111	

Command Code	Command Flags	
0x08	0x20	Set TruckID
Code (dec): 8	Flags (dec): 32	16bit Identifier: 0x0820

Sets the Truck ID of TCS3000

Modes: ALL ✖ IDLE ✖ W&M ✔ DLV ✖ BUSY ✖

COMMAND: HOST -> TCS3000 (Downlink)

Datatype:	String	
Datalength:	20 Bytes	(0 - 20)
Host data description: The Truck ID as a String (length 0 - 20 symbols)		

ANSWER: TCS3000 -> HOST (Uplink)

Datatype:	ACKNOWLEDGE	
Datalength:	0 Bytes	
Return data description: Command returns only ERRORCODE and STATUS		

Command Code	Command Flags	
0x08	0x40	Get TruckID
Code (dec): 8	Flags (dec): 64	16bit Identifier: 0x0840

Asks the Truck ID from TCS3000

Modes: ALL ✔ IDLE ✔ W&M ✔ DLV ✔ BUSY ✔

COMMAND: HOST -> TCS3000 (Downlink)

Datatype:	N/A	
Datalength:	0 Bytes	
Host data description: No data required from host		

ANSWER: TCS3000 -> HOST (Uplink)

Datatype:	String	
Datalength:	20 Bytes	(0-20)
Return data description: The Truck ID as a String (length 0 - 20 symbols)		

Command Code	Command Flags	
0x0C	0x20	Set Current Product
Code (dec): 12	Flags (dec): 32	16bit Identifier: 0x0C20

Sets the active Product ID and loads the product data TCS3000

Modes: ALL ✖ IDLE ✔ W&M ✖ DLV ✖ BUSY ✖

COMMAND: HOST -> TCS3000 (Downlink)

Datatype:	U16
Datalength:	2 Bytes
Host data description: The Product ID as U16 (Range 1001-9999)	

ANSWER: TCS3000 -> HOST (Uplink)

Datatype:	ACKNOWLEDGE
Datalength:	0 Bytes
Return data description: Command returns only ERRORCODE and STATUS	

Command Code	Command Flags	
0x0C	0x40	Get Current Product ID
Code (dec): 12	Flags (dec): 64	16bit Identifier: 0x0C40

Asks the Current Product ID from TCS3000

Modes: ALL ✔ IDLE ✔ W&M ✔ DLV ✔ BUSY ✔

COMMAND: HOST -> TCS3000 (Downlink)

Datatype:	N/A
Datalength:	0 Bytes
Host data description: No data required from host	

ANSWER: TCS3000 -> HOST (Uplink)

Datatype:	String
Datalength:	2 Bytes
Return data description: The Product ID as U16	

CONFIDENTIAL

Command Code 0x0D Code (dec): 13	Command Flags 0x20 Flags (dec): 32	Set Product Name 16bit Identifier: 0x0D20
Sets the Name of the currently active product Modes: ALL ✗ IDLE ✓ W&M ✓ DLV ✗ BUSY ✗ COMMAND: HOST -> TCS3000 (Downlink)		
Datatype: String Datalength: 20 Bytes (0 - 20) Host data description: The Product Name as a String (length 0 - 20 symbols)		
ANSWER: TCS3000 -> HOST (Uplink)		
Datatype: ACKNOWLEDGE Datalength: 0 Bytes Return data description: Command returns only ERRORCODE and STATUS		
Command Code 0x0D Code (dec): 13	Command Flags 0x40 Flags (dec): 64	Get Product Name 16bit Identifier: 0x0D40
Asks the Name of the currently active product Modes: ALL ✓ IDLE ✓ W&M ✓ DLV ✓ BUSY ✓ COMMAND: HOST -> TCS3000 (Downlink)		
Datatype: N/A Datalength: 0 Bytes Host data description: No data required from host		
ANSWER: TCS3000 -> HOST (Uplink)		
Datatype: String Datalength: 20 Bytes (0-20) Return data description: The Product Name as a String (length 0 - 20 symbols)		

CONFIDENTIAL

Command Code

0x0E

Code (dec): 14

Command Flags

0x40

Flags (dec): 64

Get Product Parameter

16bit Identifier: 0x0E40

Gets the selected parameter of the CURRENT Product from TCS3000

Modes: ALL ✗ IDLE ✓ W&M ✓ DLV ✗ BUSY ✗

COMMAND: HOST -> TCS3000 (Downlink)

Datatype:

ENUM

Datalength:

1 Bytes

Host data description: The requested parameter as ENUM(RI_PPAR_G):
See description below

ANSWER: TCS3000 -> HOST (Uplink)

Datatype:

String

Datalength:

120 Bytes

(0-120)

Return data description: The product parameter as String (0-120 characters)
Notes:
DLVGROSS_UNIT returns number formatted as string used as ENUM(RI_PVOLMETRICS) see below
DLV_AVGTEMP_UNIT returns number formatted as string used as ENUM(RI_TEMPMETRICS) see below
DLV_TOTAL_UNIT returns number formatted as string used as ENUM(RI_CURRENCY) see below
DLV_MASS_UNIT returns number formatted as string used as ENUM(RI_MASSUNIT) see below
ALL Other Parameters return their data as string (decimal separator is always "." (dot))

ENUMS RELATED TO COMMAND 0x0E (0x0E40)

RI_PPAR_G		RI_PVOLMETRICS		RI_TEMPMETRICS	
0=	ERROR	0=	ERROR	0=	ERROR
1=	DLV_GROSS_UNIT	1=	GAL	1=	F
2=	DLV_AVGTEMP_UNIT	2=	L	2=	C
3=	DLV_TOTAL_UNIT	3=	UKG	3=	K
4=	DLV_MASS_UNIT	4=	daL		
5=	DLV_PRODGROSS_VALUE	5=	dL		
6=	DLV_PRODNET_VALUE	6=	cL		
7=	DLV_PRODNAME_VALUE	7=	mL		
8=	DLV_UNITPRICE_VALUE	8=	m3		
9=	DLV_COMPTABLE_VALUE	9=	cm3		
10=	DLV_COMPTABLE_PARAM	10=	bbl		
11=	DLV_DENSITY_VALUE	11=	floz		
12=	DLV_DENSITY_UNIT	12=	ft3		
13=	DLV_TAX1_NAME	13=	in3		
14=	DLV_TAX1_VALUE				
15=	DLV_TAX1_TYPE				
16=	DLV_TAX1_UNIT				
17=	DLV_TAX2_NAME				
18=	DLV_TAX2_VALUE				
19=	DLV_TAX2_TYPE				
20=	DLV_TAX2_UNIT				
21=	AE_HOLD				
22=	DLV_PRESET_TIMING				
23=	DLV_STARTUP_VOL				
24=	DLV_VALVES				
25=	PROD_VALVE1FLOW				
26=	VALVE1_REACTPULSES				
27=	VALVE1_DELAY				
28=	VALVE1_RUN_ALONE				
29=	VALVE1_REACTCHECKADVANCE				
30=	REACTCHECK_HOLD				
31=	VALVE1_REACTCHECK				
32=	DLV_PUMP				
33=	PUMP_START_ADVANCE				
34=	PUMP_STARTER_RUN				
35=	PUMP_START_DELAY				
36=	TEMPPROBE				
37=	ZFLOWV				

Command Code

0x0F

Code (dec): 15

Command Flags

0x20

Flags (dec): 32

Set Product Parameter

16bit Identifier: 0x0F20

Sets the selected parameter of the CURRENT Product of TCS3000

Modes: ALL ✗ IDLE ✓ W&M ✓ DLV ✗ BUSY ✗

COMMAND: HOST -> TCS3000 (Downlink)

Datatype: String

Datalength: 120 Bytes

Combined ENUM and String data (1-120)

Host data description: <ENUM(RI_PPAR_S)><STRING>

Notes: See below description of the ENUMS for this command

DLV_AVGTEMP_UNIT expects number formatted as string used as ENUM(RI_TEMPMETRICS) see below

DLV_TOTAL_UNIT expects number formatted as string used as ENUM(RI_CURRENCY) see below

ALL Other Parameters expect their data as string (decimal separator is always "." (dot))

ANSWER: TCS3000 -> HOST (Uplink)

Datatype: ACKNOWLEDGE

Datalength: 0 Bytes

Return data description: Command returns only ERRORCODE and STATUS

ENUMS RELATED TO COMMAND 0x0F (0x0F20)

RI_PPAR_S	RI_CURRENCY	RI_TEMPMETRICS
0= ERROR	0= ERROR	0= ERROR
1= DLV_AVGTEMP_UNIT	1= \$	1= F
2= DLV_TOTAL_UNIT	2= USD	2= C
3= DLV_RODNAME_VALUE	3= EUR	3= K
4= DLV_UNITPRICE_VALUE	4= GBP	
5= DLV_TAX1_NAME	5= CAD	
6= DLV_TAX1_VALUE	6= MXN	
7= DLV_TAX1_TYPE	7= CLP	
8= DLV_TAX1_UNIT	8= JPY	
9= DLV_TAX2_NAME	9= BGN	
10= DLV_TAX2_VALUE	10= CHF	
11= DLV_TAX2_TYPE	11= CZK	
12= DLV_TAX2_UNIT	12= DKK	
13= AE_HOLD	13= HUF	
14= DLV_PRESET_TIMING	14= LVL	
15= DLV_STARTUP_VOL	15= LTL	
16= DLV_VALVES	16= PLN	
17= PROD_VALVE1FLOW	17= RON	
18= VALVE1_REACTPULSES	18= SEK	
19= VALVE1_DELAY	19= BRL	
20= VALVE1_RUN_ALONE	20=	
21= VALVE1_REACTCHECKADVANCE		
22= REACTCHECK_HOLD		
23= VALVE1_REACTCHECK		
24= DLV_PUMP		
25= PUMP_START_ADVANCE		
26= PUMP_STARTER_RUN		
27= PUMP_START_DELAY		
28= PUMP_START_DELAY		

Command Code 0x10 Code (dec): 16	Command Flags 0x40 Flags (dec): 64	Get Product GROSS Totalizer 16bit Identifier: 0x1040
Gets the value of the GROSS Totalizer of a product from TCS3000 Modes: ALL ✖ IDLE ✔ W&M ✔ DLV ✔ BUSY ✖ COMMAND: HOST -> TCS3000 (Downlink)		
Datatype:U16 Datalength:2 Bytes 2 or 0 Host data description: Product ID (U16) In case the value is 0 or there is no value sent the command returns the CURRENT value for the CURRENT product, otherwise returns the last stored value.		
ANSWER: TCS3000 -> HOST (Uplink)		
Datatype:Dbl Datalength:8 Bytes (0-20) Return data description: The product GROSS totalizer as Dbl		
Command Code 0x11 Code (dec): 17	Command Flags 0x40 Flags (dec): 64	Get Product NET Totalizer 16bit Identifier: 0x1140
Gets the value of the NET Totalizer of a product from TCS3000 Modes: ALL ✖ IDLE ✔ W&M ✔ DLV ✔ BUSY ✖ COMMAND: HOST -> TCS3000 (Downlink)		
Datatype:U16 Datalength:2 Bytes 2 or 0 Host data description: Product ID (U16) In case the value is 0 or there is no value sent the command returns the CURRENT value for the CURRENT product, otherwise returns the last stored value.		
ANSWER: TCS3000 -> HOST (Uplink)		
Datatype:Dbl Datalength:8 Bytes (0-20) Return data description: The product GROSS totalizer as Dbl		

Command Code	Command Flags	
0x15	0x20	<u>Set Meter Make</u>
Code (dec): 21	Flags (dec): 32	16bit Identifier: 0x1520

Sets the Meter Make string in TCS3000

Modes:	ALL ✖	IDLE ✖	W&M ✔	DLV ✖	BUSY ✖
COMMAND: HOST -> TCS3000 (Downlink)					
Datatype:	String				
Datalength:	20 Bytes	(0 - 20)			
Host data description: The Meter Make as a String (length 0 - 20 symbols)					
ANSWER: TCS3000 -> HOST (Uplink)					
Datatype:	ACKNOWLEDGE				
Datalength:	0 Bytes				
Return data description: Command returns only ERRORCODE and STATUS					

Command Code	Command Flags	
0x15	0x40	<u>Get Meter Make</u>
Code (dec): 21	Flags (dec): 64	16bit Identifier: 0x1540

Asks the Meter Make string from TCS3000

Modes:	ALL ✔	IDLE ✔	W&M ✔	DLV ✔	BUSY ✔
COMMAND: HOST -> TCS3000 (Downlink)					
Datatype:	N/A				
Datalength:	0 Bytes				
Host data description: No data required from host					
ANSWER: TCS3000 -> HOST (Uplink)					
Datatype:	String				
Datalength:	20 Bytes	(0-20)			
Return data description: The Meter Make as a String (length 0 - 20 symbols)					

CONFIDENTIAL

Command Code	Command Flags	
0x16	0x20	Set Meter Model
Code (dec): 22	Flags (dec): 32	16bit Identifier: 0x1620

Sets the Meter Model string in TCS3000

Modes:	ALL ✖	IDLE ✖	W&M ✔	DLV ✖	BUSY ✖
COMMAND: HOST -> TCS3000 (Downlink)					
Datatype:	String				
Datalength:	20 Bytes	(0 - 20)			
Host data description: The Meter Model as a String (length 0 - 20 symbols)					
ANSWER: TCS3000 -> HOST (Uplink)					
Datatype:	ACKNOWLEDGE				
Datalength:	0 Bytes				
Return data description: Command returns only ERRORCODE and STATUS					

Command Code	Command Flags	
0x16	0x40	Get Meter Model
Code (dec): 22	Flags (dec): 64	16bit Identifier: 0x1640

Asks the Meter Model string from TCS3000

Modes:	ALL ✔	IDLE ✔	W&M ✔	DLV ✔	BUSY ✔
COMMAND: HOST -> TCS3000 (Downlink)					
Datatype:	N/A				
Datalength:	0 Bytes				
Host data description: No data required from host					
ANSWER: TCS3000 -> HOST (Uplink)					
Datatype:	String				
Datalength:	20 Bytes	(0-20)			
Return data description: The Meter Model as a String (length 0 - 20 symbols)					

CONFIDENTIAL

Command Code	Command Flags
0x17	0x20
Code (dec): 23	Flags (dec): 32

Set Meter Serial

16bit Identifier: 0x1720

Sets the Meter Serial Number string in TCS3000

Modes: ALL ✗ IDLE ✗ W&M ✓ DLV ✗ BUSY ✗

COMMAND: HOST -> TCS3000 (Downlink)

Datatype:	String
Datalength:	20 Bytes (0 - 20)
Host data description: The Meter Serial Number as a String (length 0 - 20 symbols)	

ANSWER: TCS3000 -> HOST (Uplink)

Datatype:	ACKNOWLEDGE
Datalength:	0 Bytes
Return data description: Command returns only ERRORCODE and STATUS	

Command Code	Command Flags
0x17	0x40
Code (dec): 23	Flags (dec): 64

Get Meter Serial

16bit Identifier: 0x1740

Asks the Meter Serial Number string from TCS3000

Modes: ALL ✓ IDLE ✓ W&M ✓ DLV ✓ BUSY ✓

COMMAND: HOST -> TCS3000 (Downlink)

Datatype:	N/A
Datalength:	0 Bytes
Host data description: No data required from host	

ANSWER: TCS3000 -> HOST (Uplink)

Datatype:	String
Datalength:	20 Bytes (0-20)
Return data description: The Meter Serial Number as a String (0 - 20 symbols)	

CONFIDENTIAL

Command Code	Command Flags	
0x19	0x40	Get CalBolt Status
Code (dec): 25	Flags (dec): 64	16bit Identifier: 0x1940
Asks the Calibration Bolt Status from TCS3000		
Modes: ALL ✓ IDLE ✓ W&M ✓ DLV ✓ BUSY ✓		
COMMAND: HOST -> TCS3000 (Downlink)		
Datatype:	N/A	
Datalength:	0 Bytes	
Host data description: No data required from host		
ANSWER: TCS3000 -> HOST (Uplink)		
Datatype:	ENUM	
Datalength:	1 Bytes	
Return data description: The CalBolt Status ENUM(RI_CALBOLT)		
0=ERROR		
1=REMOVED		
2=IN PLACE		

CONFIDENTIAL

Command Code	Command Flags	
0x1A	0x20	Set NoFlow Timer
Code (dec): 26	Flags (dec): 32	16bit Identifier: 0x1A20

Sets the No Flow Timeout in TCS3000

Modes: ALL ✗ IDLE ✗ W&M ✓ DLV ✗ BUSY ✗

COMMAND: HOST -> TCS3000 (Downlink)

Datatype:	U16
Datalength:	2 Bytes
Host data description: The no flow timeout in seconds (U16)	

ANSWER: TCS3000 -> HOST (Uplink)

Datatype:	ACKNOWLEDGE
Datalength:	0 Bytes
Return data description: Command returns only ERRORCODE and STATUS	

Command Code	Command Flags	
0x1A	0x40	Get NoFlow Timer
Code (dec): 26	Flags (dec): 64	16bit Identifier: 0x1A40

Asks the No Flow Timeout from TCS3000

Modes: ALL ✓ IDLE ✓ W&M ✓ DLV ✓ BUSY ✓

COMMAND: HOST -> TCS3000 (Downlink)

Datatype:	N/A
Datalength:	0 Bytes
Host data description: No data required from host	

ANSWER: TCS3000 -> HOST (Uplink)

Datatype:	U16
Datalength:	2 Bytes
Return data description: The no flow timeout in seconds (U16)	

CONFIDENTIAL

Command Code		Command Flags	
0x1C		0x40	
Code (dec): 28		Flags (dec): 64	

Get SysNet Totalizer

16bit Identifier: 0x1C40

Asks the System NET Totalizer value from TCS3000

Modes: ALL ✓ IDLE ✓ W&M ✓ DLV ✓ BUSY ✓

COMMAND: HOST -> TCS3000 (Downlink)

Datatype:	N/A
Datalength:	0 Bytes
Host data description: No data required from host	

ANSWER: TCS3000 -> HOST (Uplink)

Datatype:	Dbl
Datalength:	8 Bytes
Return data description: The system NET Totalizer as a Dbl	

CONFIDENTIAL

Command Code	Command Flags	
0x1D	0x20	Set Next Ticket Nr
Code (dec): 29	Flags (dec): 32	16bit Identifier: 0x1D20

Sets the Next Delivery Ticket Number in TCS3000

Modes: ALL ✗ IDLE ✓ W&M ✓ DLV ✗ BUSY ✗

COMMAND: HOST -> TCS3000 (Downlink)

Datatype:	U64
Datalength:	8 Bytes
Host data description: The next Delivery Ticket Number as U64	

ANSWER: TCS3000 -> HOST (Uplink)

Datatype:	ACKNOWLEDGE
Datalength:	0 Bytes
Return data description: Command returns only ERRORCODE and STATUS	

Command Code	Command Flags	
0x1D	0x40	Get Next Ticket Nr
Code (dec): 29	Flags (dec): 64	16bit Identifier: 0x1D40

Asks the Next Delivery Ticket Number from TCS3000

Modes: ALL ✓ IDLE ✓ W&M ✓ DLV ✓ BUSY ✓

COMMAND: HOST -> TCS3000 (Downlink)

Datatype:	N/A
Datalength:	0 Bytes
Host data description: No data required from host	

ANSWER: TCS3000 -> HOST (Uplink)

Datatype:	U64
Datalength:	8 Bytes
Return data description: The next Delivery Ticket Number as U64	

CONFIDENTIAL

Command Code		Command Flags	
0x1E		0x40	
Code (dec): 30		Flags (dec): 64	

Get SysGross Totalizer

16bit Identifier: 0x1E40

Asks the System GROSS Totalizer value from TCS3000

Modes: ALL ✓ IDLE ✓ W&M ✓ DLV ✓ BUSY ✓

COMMAND: HOST -> TCS3000 (Downlink)

Datatype:	N/A
Datalength:	0 Bytes
Host data description: No data required from host	

ANSWER: TCS3000 -> HOST (Uplink)

Datatype:	Dbl
Datalength:	8 Bytes
Return data description: The system Gross Totalizer as a Dbl	

CONFIDENTIAL

Command Code		Command Flags	
0x1F		0x40	
Code (dec): 31		Flags (dec): 64	

16bit Identifier: 0x1F40

Get Delivery State

Asks the Delivery State from TCS3000

Modes: ALL ✓ IDLE ✓ W&M ✓ DLV ✓ BUSY ✓

COMMAND: HOST -> TCS3000 (Downlink)

Datatype:	N/A
Datalength:	0 Bytes
Host data description: No data required from host	

ANSWER: TCS3000 -> HOST (Uplink)

Datatype:	ENUM
Datalength:	1 Bytes
Return data description: The system Delivery State as ENUM(DLVSTATE) 0=ERROR 1=IDLE 2=ACTIVE 3=AIR (Air detected) 4=PAUSED 5=STOPPED 6=TCKT_PENDING 7=PRINTING 8=MULTIPLE DELIVERY PENDING	

CONFIDENTIAL

Command Code	Command Flags	
0x20	0x20	Set Date Format
Code (dec): 32	Flags (dec): 32	16bit Identifier: 0x2020

Sets the System Date Format in TCS3000

Modes:	ALL ✖	IDLE ✔	W&M ✔	DLV ✖	BUSY ✖
--------	-------	--------	-------	-------	--------

COMMAND: HOST -> TCS3000 (Downlink)

Datatype:	ENUM
Datalength:	1 Bytes
Host data description: The date format as ENUM(RI_DATEFORM) 0=ERROR 1=MM/dd/yyyy 2=yy/MM/dd 3=MMMM dd, yyyy 4=dd/mm/yyyy 5=dd/mm/yy	

ANSWER: TCS3000 -> HOST (Uplink)

Datatype:	ACKNOWLEDGE
Datalength:	0 Bytes
Return data description: Command returns only ERRORCODE and STATUS	

Command Code	Command Flags	
0x20	0x40	Get Date Format
Code (dec): 32	Flags (dec): 64	16bit Identifier: 0x2040

Asks the System Date Format from TCS3000

Modes:	ALL ✔	IDLE ✔	W&M ✔	DLV ✔	BUSY ✔
--------	-------	--------	-------	-------	--------

COMMAND: HOST -> TCS3000 (Downlink)

Datatype:	N/A
Datalength:	0 Bytes
Host data description: No data required from host	

ANSWER: TCS3000 -> HOST (Uplink)

Datatype:	ENUM
Datalength:	1 Bytes
Return data description: The date format as ENUM(RI_DATEFORM) 0=ERROR 1=MM/dd/yyyy 2=yy/MM/dd 3=MMMM dd, yyyy 4=dd/mm/yyyy 5=dd/mm/yy	

Command Code

0x21

Code (dec): 33

Command Flags

0x20

Flags (dec): 32

Set Time Format

16bit Identifier: 0x2120

Sets the System Time Format in TCS3000

Modes: ALL ✗ IDLE ✓ W&M ✓ DLV ✗ BUSY ✗

COMMAND: HOST -> TCS3000 (Downlink)

Datatype:

ENUM

Datalength:

1 Bytes

Host data description: The date format as ENUM(RI_TIMEFORM)
0=ERROR
1=AM/PM
2=24 Hours

ANSWER: TCS3000 -> HOST (Uplink)

Datatype:

ACKNOWLEDGE

Datalength:

0 Bytes

Return data description: Command returns only ERRORCODE and STATUS

Command Code

0x21

Code (dec): 33

Command Flags

0x40

Flags (dec): 64

Get Time Format

16bit Identifier: 0x2140

Asks the System Time Format from TCS3000

Modes: ALL ✓ IDLE ✓ W&M ✓ DLV ✓ BUSY ✓

COMMAND: HOST -> TCS3000 (Downlink)

Datatype:

N/A

Datalength:

0 Bytes

Host data description: No data required from host

ANSWER: TCS3000 -> HOST (Uplink)

Datatype:

ENUM

Datalength:

1 Bytes

Return data description: The date format as ENUM(RI_TIMEFORM)
0=ERROR
1=AM/PM
2=24 Hours

CONFIDENTIAL

COMMAND 0x22 is not functional at the moment!

Command Code	Command Flags
0x22	0x40
Code (dec): 34	Flags (dec): 64

Get Temperature Metrics

16bit Identifier: 0x2240

N/A

Modes:

ALL ✖

IDLE ✖

W&M ✖

DLV ✖

BUSY ✖

COMMAND: HOST -> TCS3000 (Downlink)

Datatype:	N/A
Datalength:	0 Bytes
Host data description: No data required from host	

ANSWER: TCS3000 -> HOST (Uplink)

Datatype:	String
Datalength:	8 Bytes
Return data description:	

CONFIDENTIAL

Command Code

0x23

Code (dec): 35

Command Flags

0x20

Flags (dec): 32

Set Decimal Point

16bit Identifier: 0x2320

Sets the Decimal Symbol in TCS3000

Modes:

ALL

IDLE

W&M

DLV

BUSY

COMMAND: HOST -> TCS3000 (Downlink)

Datatype:

ENUM

Datalength:

1 Bytes

Host data description: The decimal symbol as ENUM(RI_DECIMAL)
0=ERROR
1=. (dot)
2=, (comma)

ANSWER: TCS3000 -> HOST (Uplink)

Datatype:

ACKNOWLEDGE

Datalength:

0 Bytes

Return data description: Command returns only ERRORCODE and STATUS

Command Code

0x23

Code (dec): 35

Command Flags

0x40

Flags (dec): 64

Get Decimal Point

16bit Identifier: 0x2340

Asks the Decimal Symbol from TCS3000

Modes:

ALL

IDLE

W&M

DLV

BUSY

COMMAND: HOST -> TCS3000 (Downlink)

Datatype:

N/A

Datalength:

0 Bytes

Host data description: No data required from host

ANSWER: TCS3000 -> HOST (Uplink)

Datatype:

ENUM

Datalength:

1 Bytes

Return data description: The decimal symbol as ENUM(RI_DECIMAL)
0=ERROR
1=. (dot)
2=, (comma)

CONFIDENTIAL

Command Code

0x27

Code (dec): 39

Command Flags

0x20

Flags (dec): 32

Set CustID Value

16bit Identifier: 0x2720

Sets a Customer ID value in TCS3000

Modes: ALL ✖ IDLE ✔ W&M ✖ DLV ✖ BUSY ✖

COMMAND: HOST -> TCS3000 (Downlink)

Datatype: String

ENUM(RI_CUSTIDNR);StringValue

Datalength: 120 Bytes

1-120

Host data description: Combined String data: <ENUM><String>
ENUM(RI_CUSTIDNR) values:
0=ERROR
1=CUSTID1
2=CUSTID2
3=CUSTID3
4=CUSTID4

ANSWER: TCS3000 -> HOST (Uplink)

Datatype: ACKNOWLEDGE

Datalength: 0 Bytes

Return data description: Command returns only ERRORCODE and STATUS

Command Code

0x28

Code (dec): 40

Command Flags

0x40

Flags (dec): 64

Get CustID Value

16bit Identifier: 0x2840

Gets a Customer ID value from TCS3000

Modes: ALL ✖ IDLE ✔ W&M ✖ DLV ✔ BUSY ✖

COMMAND: HOST -> TCS3000 (Downlink)

Datatype: ENUM

Datalength: 1 Bytes

Host data description: ENUM(RI_CUSTIDNR) values:
0=ERROR
1=CUSTID1
2=CUSTID2
3=CUSTID3
4=CUSTID4

ANSWER: TCS3000 -> HOST (Uplink)

Datatype: String

Datalength: 120 Bytes

1-120

Return data description: The string value of the selected customer ID

CONFIDENTIAL

Command Code	Command Flags	
0x29	0x20	<u>Set CustID Label</u>
Code (dec): 41	Flags (dec): 32	16bit Identifier: 0x2920

Sets a Customer ID Label value in TCS3000

Modes: ALL ☐ IDLE ☒ W&M ☐ DLV ☐ BUSY ☐

COMMAND: HOST -> TCS3000 (Downlink)

Datatype:	String	ENUM(RI_CUSTIDNR);StringValue
Datalength:	120 Bytes	1-120
Host data description: Combined String data: <ENUM><String> ENUM(RI_CUSTIDNR) values: 0=ERROR 1=CUSTID1 2=CUSTID2 3=CUSTID3 4=CUSTID4		

ANSWER: TCS3000 -> HOST (Uplink)

Datatype:	ACKNOWLEDGE
Datalength:	0 Bytes
Return data description: Command returns only ERRORCODE and STATUS	

Command Code	Command Flags	
0x2A	0x40	<u>Get CustID Label</u>
Code (dec): 42	Flags (dec): 64	16bit Identifier: 0x2A40

Gets a Customer ID Label value from TCS3000

Modes: ALL ☐ IDLE ☒ W&M ☐ DLV ☒ BUSY ☐

COMMAND: HOST -> TCS3000 (Downlink)

Datatype:	ENUM
Datalength:	1 Bytes
Host data description: ENUM(RI_CUSTIDNR) values: 0=ERROR 1=CUSTID1 2=CUSTID2 3=CUSTID3 4=CUSTID4	

ANSWER: TCS3000 -> HOST (Uplink)

Datatype:	String
Datalength:	120 Bytes
Return data description: The string value of the selected customer ID	

Command Code	Command Flags	
0x2B	0x40	<u>Get Gross Display</u>
Code (dec): 43	Flags (dec): 64	16bit Identifier: 0x2B40

Asks the Delivery GROSS value from TCS3000

Modes: ALL ✓ IDLE ✓ W&M ✓ DLV ✓ BUSY ✓

COMMAND: HOST -> TCS3000 (Downlink)

Datatype:	N/A
Datalength:	0 Bytes
Host data description: No data required from host	

ANSWER: TCS3000 -> HOST (Uplink)

Datatype:	Dbl
Datalength:	8 Bytes
Return data description: The Delivery Gross as a Dbl	

Command Code	Command Flags	
0x2C	0x40	<u>Get Net Display</u>
Code (dec): 44	Flags (dec): 64	16bit Identifier: 0x2C40

Asks the Delivery NET value from TCS3000

Modes: ALL ✓ IDLE ✓ W&M ✓ DLV ✓ BUSY ✓

COMMAND: HOST -> TCS3000 (Downlink)

Datatype:	N/A
Datalength:	0 Bytes
Host data description: No data required from host	

ANSWER: TCS3000 -> HOST (Uplink)

Datatype:	Dbl
Datalength:	8 Bytes
Return data description: The Delivery Net as a Dbl	

Command Code	Command Flags	
0x2D	0x40	<u>Get Volume Display</u>
Code (dec): 45	Flags (dec): 64	16bit Identifier: 0x2D40

Asks the Delivery Volume (automatic Gross or Net) value from TCS3000

Modes: ALL ✓ IDLE ✓ W&M ✓ DLV ✓ BUSY ✓

COMMAND: HOST -> TCS3000 (Downlink)

Datatype:	N/A
Datalength:	0 Bytes
Host data description: No data required from host	

ANSWER: TCS3000 -> HOST (Uplink)

Datatype:	Dbl
Datalength:	8 Bytes
Return data description: The Delivery Volume as a Dbl Depending on the delivery and product if NET is enabled for the current/last delivery this command returns NET volume otherwise it returns Gross	

Command Code	Command Flags	
0x30	0x20	Set Current Temp
Code (dec): 48	Flags (dec): 32	16bit Identifier: 0x3020
Calibrates the offset of the Temperature probe of TCS3000		
Modes: ALL ✗ IDLE ✗ W&M ✓ DLV ✗ BUSY ✗		
COMMAND: HOST -> TCS3000 (Downlink)		
Datatype:	String	ENUM(RI_TEMPMETRICS);Dbl
Datalength:	9 Bytes	
Host data description: Combined String data: <ENUM><Dbl> Dbl - Current reading from external reference sensor ENUM(RI_TEMPMETRICS) - The unit of the reference sensor: 0=ERROR 1=F 2=C 3=K		
ANSWER: TCS3000 -> HOST (Uplink)		
Datatype:	ACKNOWLEDGE	
Datalength:	0 Bytes	
Return data description: Command returns only ERRORCODE and STATUS		
Command Code	Command Flags	
0x30	0x40	Get Current Temp
Code (dec): 48	Flags (dec): 64	16bit Identifier: 0x3040
Asks the current temperature measurement value from TCS3000		
Modes: ALL ✗ IDLE ✓ W&M ✓ DLV ✗ BUSY ✗		
COMMAND: HOST -> TCS3000 (Downlink)		
Datatype:	N/A	
Datalength:	0 Bytes	
Host data description: No data required from host		
ANSWER: TCS3000 -> HOST (Uplink)		
Datatype:	Dbl	
Datalength:	8 Bytes	
Return data description: Current Temperature as a Dbl in the last selected T Unit		

Command Code

0x35

Code (dec): 53

Command Flags

0x40

Flags (dec): 64

Get Shift Status

16bit Identifier: 0x3540

Asks the status of the Shift from TCS3000

Modes: ALL ✓ IDLE ✓ W&M ✓ DLV ✓ BUSY ✓

COMMAND: HOST -> TCS3000 (Downlink)

Datatype:

N/A

Datalength:

0 Bytes

Host data description: No data required from host

ANSWER: TCS3000 -> HOST (Uplink)

Datatype:

ENUM

Datalength:

1 Bytes

Return data description: The shift status as ENUM(RI_SHIFT_STATE)
0=ERROR
1=NOT_ACTIVE
2=ACTIVE

Command Code

0x36

Code (dec): 54

Command Flags

0x20

Flags (dec): 32

Cancel Configured Delivery

16bit Identifier: 0x3620

Cancels the configuration process for already configured (but not started) delivery

Modes: ALL ✗ IDLE ✓ W&M ✗ DLV ✗ BUSY ✗

COMMAND: HOST -> TCS3000 (Downlink)

Datatype:

N/A

Datalength:

0 Bytes

Host data description: No data required from host

ANSWER: TCS3000 -> HOST (Uplink)

Datatype:

ACKNOWLEDGE

Datalength:

0 Bytes

Return data description: Command returns only ERROR CODE and STATUS

CONFIDENTIAL

Command Code		Command Flags		Configure Direct Delivery
0x37		0x20		
Code (dec): 55		Flags (dec): 32		
16bit Identifier: 0x3720				
Configures a DIRECT Delivery based on product ID				
Modes: ALL ✗ IDLE ✓ W&M ✗ DLV ✗ BUSY ✗				
COMMAND: HOST -> TCS3000 (Downlink)				
Datatype:		U16		
Datalength:		2 Bytes		
Host data description: Product ID (U16) Range 1001-9999				
ANSWER: TCS3000 -> HOST (Uplink)				
Datatype:		ACKNOWLEDGE		
Datalength:		0 Bytes		
Return data description: Command returns only ERRORCODE and STATUS				
Command Code		Command Flags		Configure Preset Delivery
0x38		0x20		
Code (dec): 56		Flags (dec): 32		
16bit Identifier: 0x3820				
Configures a PRESET Delivery				
Modes: ALL ✗ IDLE ✓ W&M ✗ DLV ✗ BUSY ✗				
COMMAND: HOST -> TCS3000 (Downlink)				
Datatype:		String		DivType(ENUM),ProdId(U16),PresetAmount(Dbl)
Datalength:		11 Bytes		DivType -> ENUM(RI_DLVTYPE)
Host data description: Combined Numeric data: <ENUM><U16><Dbl> <ENUM(RI_DLVTYPE)><PRODID(U16)><Amount(Dbl)> Product ID: (U16) Range 1001-9999 Amount: Dbl ENUM(RI_DLVTYPE): 0=ERROR 1=DIRECT (Direct Delivery) 2=PRESET_NET_VOLUME (Net Volume Preset) 3=PRESET_GRS_VOLUME (Gross Volume Preset) 4=PRESET_AUTO_VOLUME (Auto (Net/Gross) Volume Preset) 5=PRESET_NET_PRICE (Net Price Preset) 6=PRESET_GRS_PRICE (Gross Price Preset)				
ANSWER: TCS3000 -> HOST (Uplink)				
Datatype:		ACKNOWLEDGE		
Datalength:		0 Bytes		
Return data description: Command returns only ERRORCODE and STATUS				

Command Code	Command Flags	
0x39	0x20	<u>Pause Delivery</u>
Code (dec): 57	Flags (dec): 32	16bit Identifier: 0x3920

Pauses already started delivery

Modes:	ALL ✖	IDLE ✖	W&M ✖	DLV ✔	BUSY ✖
--------	-------	--------	-------	-------	--------

COMMAND: HOST -> TCS3000 (Downlink)

Datatype:	N/A
Datalength:	0 Bytes
Host data description: No data required from host	

ANSWER: TCS3000 -> HOST (Uplink)

Datatype:	ACKNOWLEDGE
Datalength:	0 Bytes
Return data description: Command returns only ERRORCODE and STATUS	

Command Code	Command Flags	
0x3A	0x20	<u>Resume Delivery</u>
Code (dec): 58	Flags (dec): 32	16bit Identifier: 0x3A20

Resumes already started delivery that was paused

Modes:	ALL ✖	IDLE ✖	W&M ✖	DLV ✔	BUSY ✖
--------	-------	--------	-------	-------	--------

COMMAND: HOST -> TCS3000 (Downlink)

Datatype:	N/A
Datalength:	0 Bytes
Host data description: No data required from host	

ANSWER: TCS3000 -> HOST (Uplink)

Datatype:	ACKNOWLEDGE
Datalength:	0 Bytes
Return data description: Command returns only ERRORCODE and STATUS	

Command Code	Command Flags	
0x3B	0x20	<u>Abort Delivery</u>
Code (dec): 59	Flags (dec): 32	16bit Identifier: 0x3B20

Aborts already started delivery - WITHOUT Waiting for normal delivery end

Modes:	ALL ✖	IDLE ✖	W&M ✖	DLV ✔	BUSY ✖
--------	-------	--------	-------	-------	--------

COMMAND: HOST -> TCS3000 (Downlink)

Datatype:	N/A
Datalength:	0 Bytes
Host data description: No data required from host	

ANSWER: TCS3000 -> HOST (Uplink)

Datatype:	ACKNOWLEDGE
Datalength:	0 Bytes
Return data description: Command returns only ERRORCODE and STATUS	

Command Code 0x3C Code (dec): 60	Command Flags 0x20 Flags (dec): 32	Begin Delivery 16bit Identifier: 0x3C20
Starts already configured delivery Note: The ENUM value of this command MUST match the configured type of delivery! Modes: ALL ☐ IDLE ☒ W&M ☐ DLV ☐ BUSY ☐ COMMAND: HOST -> TCS3000 (Downlink)		
Datatype: ENUM Datalength: 1 Bytes	DivType -> ENUM(RI_DLVTYPE) Host data description: ENUM(RI_DVTYPE): 0=ERROR 1=DIRECT (Direct Delivery) 2=PRESET_NET (Net Volume Preset) 3=PRESET_GRS (Gross Volume Preset) 4=PRESET_AUTO (Auto (Net/Gross) Volume Preset) 5=PRESET_PRS_NET (Gross Price Preset) 6=PRESET_PRS_GRS (Net Price Preset)	
ANSWER: TCS3000 -> HOST (Uplink)		
Datatype: ACKNOWLEDGE Datalength: 0 Bytes	Return data description: Command returns only ERRORCODE and STATUS	
Command Code 0x3D Code (dec): 61	Command Flags 0x20 Flags (dec): 32	End Delivery 16bit Identifier: 0x3D20
Finalizes already started delivery Used mainly with Direct delivery, not needed when Preset delivery has finished normally Can be used to interrupt normally a Preset delivery Modes: ALL ☐ IDLE ☐ W&M ☐ DLV ☒ BUSY ☐ COMMAND: HOST -> TCS3000 (Downlink)		
Datatype: N/A Datalength: 0 Bytes	Host data description: No data required from host	
ANSWER: TCS3000 -> HOST (Uplink)		
Datatype: ACKNOWLEDGE Datalength: 0 Bytes	Return data description: Command returns only ERRORCODE and STATUS	

Command Code

0x3E

Code (dec): 62

Command Flags

0x20

Flags (dec): 32

Print Internal Ticket

16bit Identifier: 0x3E20

Prints the internal ticket for the last delivery (same as the print button)
Note - this command also generates the Internal G-Ticket

Modes:

ALL ✖

IDLE ✔

W&M ✖

DLV ✖

BUSY ✖

COMMAND: HOST -> TCS3000 (Downlink)

Datatype:

N/A

Datalength:

0 Bytes

Host data description: No data required from host

ANSWER: TCS3000 -> HOST (Uplink)

Datatype:

ACKNOWLEDGE

Datalength:

0 Bytes

Return data description: Command returns only ERRORCODE and STATUS

Command Code

0x3F

Code (dec): 63

Command Flags

0x20

Flags (dec): 32

Generate Internal Ticket

16bit Identifier: 0x3F20

Generates the internal ticket for the last delivery and saves it in the G-Ticket
Note - this command only generates the ticket without sending it to the printer!

Modes:

ALL ✖

IDLE ✔

W&M ✖

DLV ✖

BUSY ✖

COMMAND: HOST -> TCS3000 (Downlink)

Datatype:

N/A

Datalength:

0 Bytes

Host data description: No data required from host

ANSWER: TCS3000 -> HOST (Uplink)

Datatype:

ACKNOWLEDGE

Datalength:

0 Bytes

Return data description: Command returns only ERRORCODE and STATUS

CONFIDENTIAL

Command Code

0x40

Code (dec): 64

Command Flags

0x40

Flags (dec): 64

Get Average Temperature

16bit Identifier: 0x4040

Gets the value of the average temperature of the currently active (or the previous) delivery

Modes: ALL ✓ IDLE ✓ W&M ✓ DLV ✓ BUSY ✓

COMMAND: HOST -> TCS3000 (Downlink)

Datatype:

N/A

Datalength:

0 Bytes

Host data description: No data required from host

ANSWER: TCS3000 -> HOST (Uplink)

Datatype:

Dbl

Datalength:

8 Bytes

Return data description: The Average Temperature as a Dbl in the T Unit of the product

Command Code

0x42

Code (dec): 66

Command Flags

0x40

Flags (dec): 64

Get Flowrate

16bit Identifier: 0x4240

Gets the current value of the flowrate

Modes: ALL ✓ IDLE ✓ W&M ✓ DLV ✓ BUSY ✓

COMMAND: HOST -> TCS3000 (Downlink)

Datatype:

N/A

Datalength:

0 Bytes

Host data description: No data required from host

ANSWER: TCS3000 -> HOST (Uplink)

Datatype:

Dbl

Datalength:

8 Bytes

Return data description: The current value of the flowrate in the volume unit of the product

Command Code	Command Flags	
0x43	0x20	<u>Reset No Flow Timer</u>
Code (dec): 67	Flags (dec): 32	16bit Identifier: 0x4320

Resets the Delivery No Flow Timeout Timer

Modes:	ALL ✖	IDLE ✖	W&M ✖	DLV ✔	BUSY ✖
COMMAND: HOST -> TCS3000 (Downlink)					
Datatype:	N/A				
Datalength:	0 Bytes				
Host data description: No data required from host					
ANSWER: TCS3000 -> HOST (Uplink)					
Datatype:	ACK				
Datalength:	0 Bytes				
Return data description: Command returns only ERRORCODE and STATUS					

Command Code	Command Flags	
0x43	0x40	<u>Get No Flow Timer Remail</u>
Code (dec): 67	Flags (dec): 64	16bit Identifier: 0x4340

Gets the remaining time until No Flow Timer Expires (in ms)

Modes:	ALL ✖	IDLE ✖	W&M ✖	DLV ✔	BUSY ✖
COMMAND: HOST -> TCS3000 (Downlink)					
Datatype:	N/A				
Datalength:	0 Bytes				
Host data description: No data required from host					
ANSWER: TCS3000 -> HOST (Uplink)					
Datatype:	U32				
Datalength:	4 Bytes				
Return data description: The remaining time until No Flow Timer expires in ms					

CONFIDENTIAL

Command Code 0x4E Code (dec): 78	Command Flags 0x40 Flags (dec): 64	Get Product Status 16bit Identifier: 0x4E40
Gets the status of the requested product (by requested ProdID)		
Modes: ALL ✓ IDLE ✓ W&M ✓ DLV ✓ BUSY ✓		
COMMAND: HOST -> TCS3000 (Downlink)		
Datatype: U16	Datalength: 2 Bytes	
Host data description: Product ID U16 (Range 1001-9999)		
ANSWER: TCS3000 -> HOST (Uplink)		
Datatype: ENUM	Datalength: 1 Bytes	
Return data description: Returns the status as ENUM(RI_PRODSTAT) 0=ERROR or NOT CALIBRATED 1=CALIBRATED 2=ERROR 3=ACTIVE 4=ACTIVE & SELECTED		
Command Code 0x4F Code (dec): 79	Command Flags 0x40 Flags (dec): 64	Get Printer Status 16bit Identifier: 0x4F40
Gets the TCS3000 Printer Status		
Modes: ALL ✓ IDLE ✓ W&M ✓ DLV ✓ BUSY ✓		
COMMAND: HOST -> TCS3000 (Downlink)		
Datatype: N/A	Datalength: 0 Bytes	
Host data description: No data required from host		
ANSWER: TCS3000 -> HOST (Uplink)		
Datatype: ENUM	Datalength: 1 Bytes	
Return data description: Returns the status of the printer as ENUM(PRN_STATUS) 0=N/A 1=OK 2=PAPER 3=PRINTER ERROR 4=BUSY		

Command Code

0x54

Code (dec): 84

Command Flags

0x40

Flags (dec): 64

Get Number of E-Tickets

16bit Identifier: 0x5440

Returns the total number of E-Tickets from a particular type currently stored inside the TCS3000

Note - this command generates internal table inside the TCS3000 used for the commands below

Modes: ALL ✖ IDLE ✔ W&M ✔ DLV ✖ BUSY ✖

COMMAND: HOST -> TCS3000 (Downlink)

Datatype:	ENUM
Datalength:	1 Bytes
Host data description: E-Ticket Type as ENUM(RI_ETKT_TYPE) 0=ERROR 1=NONPRINTED (Delivery Tickets only) 2=PRINTED (But not transferred Delivery Tickets only) 3=TRANSFERRED (Delivery Tickets only) 4=ALL_DLVTICKETS (Applies only to Delivery Tickets) 5=SHIFT 6=PROVER	

ANSWER: TCS3000 -> HOST (Uplink)

Datatype:	U32
Datalength:	4 Bytes
Return data description: The number of E-Tickets from the selected type stored inside the TCS3000	

CONFIDENTIAL

Command Code	Command Flags
0x55	0x40
Code (dec): 85	Flags (dec): 64

Get First E-Ticket Number

16bit Identifier: 0x5540

Gets the **First** E-Ticket Number from the table generated by command 0x54
Note - you need to run command 0x54 at least once before this command every time you change the E-Ticket type

Modes: ALL ✖ IDLE ✔ W&M ✔ DLV ✖ BUSY ✖

COMMAND: HOST -> TCS3000 (Downlink)

Datatype:	N/A
Datalength:	0 Bytes
Host data description: No data required from host	

ANSWER: TCS3000 -> HOST (Uplink)

Datatype:	U64
Datalength:	8 Bytes
Return data description: Returns the E-Ticket NR (E-Ticket ID) as U64	

Command Code	Command Flags
0x56	0x40
Code (dec): 86	Flags (dec): 64

Get Next E-Ticket Number

16bit Identifier: 0x5640

Gets the **Next** E-Ticket Number from the table generated by command 0x54
Note - if you run this command right after command 0x54 it will return the **FIRST** E-Ticket number
Note - you need to run command 0x54 at least once before this command every time you change the E-Ticket type

Modes: ALL ✖ IDLE ✔ W&M ✔ DLV ✖ BUSY ✖

COMMAND: HOST -> TCS3000 (Downlink)

Datatype:	N/A
Datalength:	0 Bytes
Host data description: No data required from host	

ANSWER: TCS3000 -> HOST (Uplink)

Datatype:	U64
Datalength:	8 Bytes
Return data description: Returns the E-Ticket NR (E-Ticket ID) as U64	

Command Code	Command Flags
0x57	0x40
Code (dec): 87	Flags (dec): 64

Get E-Ticket Number

16bit Identifier: 0x5740

Gets the E-Ticket Number specified by the index from the table generated by command 0x54
Note - you need to run command 0x54 at least once before this command every time you change the E-Ticket type
Note - it is recommended to use this command when downloading the whole E-tickets table

Modes: ALL ✖ IDLE ✔ W&M ✔ DLV ✖ BUSY ✖

COMMAND: HOST -> TCS3000 (Downlink)

Datatype:	U32
Datalength:	4 Bytes
Host data description: Index of the element (0 based) in the table generated by command 0x54	

ANSWER: TCS3000 -> HOST (Uplink)

Datatype:	U64
Datalength:	8 Bytes
Return data description: Returns the requested E-Ticket NR (E-Ticket ID) as U64	

Command Code	Command Flags	
0x59	0x20	<u>Clear Print Status E-Ticket NR</u>
Code (dec): 89	Flags (dec): 32	16bit Identifier: 0x5920
Clears the print status of the specified ticket - sets it to PRINTED and moves the ticket type PRINTED Note - if the ticket specified was already marked PRINTED or TRANSFERRED no changes are made to it Note - This command is applicable ONLY to Delivery tickets		
Modes: ALL ✖ IDLE ✔ W&M ✔ DLV ✖ BUSY ✖		
COMMAND: HOST -> TCS3000 (Downlink)		
Datatype:	U64	
Datalength:	8 Bytes	
Host data description: The selected E-Ticket NR (E-Ticket ID)		
ANSWER: TCS3000 -> HOST (Uplink)		
Datatype:	ACKNOWLEDGE	
Datalength:	0 Bytes	
Return data description: Command returns only ERRORCODE and STATUS		

CONFIDENTIAL

Command Code

0x5A

Code (dec): 90

Command Flags

0x40

Flags (dec): 64

ROPEN E-Ticket

16bit Identifier: 0x5A40

Opens the selected E-Ticket in the memory of TCS3000 and returns the number of datablocks in it

Modes:

ALL ✖

IDLE ✔

W&M ✔

DLV ✖

BUSY ✖

COMMAND: HOST -> TCS3000 (Downlink)

Datatype:	String	TicketType(ENUM),E-TicketNR(U64)
Datalength:	9 Bytes	
Host data description: Combined Numeric data: <ENUM><U64> <ENUM(RI_ETKT_TYPE)><E-TicketNR(U64)> ENUM(RI_ETKT_TYPE): 0=ERROR 1=NONPRINTED (Delivery Tickets only) 2=PRINTED (But not transferred Delivery Tickets only) 3=TRANSFERRED (Delivery Tickets only) 4=ALL_DLV TICKETS (Applies only to Delivery Tickets) 5=SHIFT 6=PROVER		

ANSWER: TCS3000 -> HOST (Uplink)

Datatype:	U32
Datalength:	4 Bytes
Return data description: Number of datablocks for transferring the ticket	

CONFIDENTIAL

Command Code	Command Flags
0x5B	0x40
Code (dec): 91	Flags (dec): 64

Get E-Ticket FIRST Block

16bit Identifier: 0x5B40

Gets the FIST Datablock from already open E-Ticket
Note - Before running this command you must open (for reading) an E-Ticket in the memory of the TCS3000
Note - The length of all datablocks during E-Ticket transfer is the same (usually 120 symbols) except for the last one

Modes: ALL ✖ IDLE ✔ W&M ✔ DLV ✖ BUSY ✖

COMMAND: HOST -> TCS3000 (Downlink)

Datatype:	N/A
Datalength:	0 Bytes
Host data description: No data required from host	

ANSWER: TCS3000 -> HOST (Uplink)

Datatype:	String
Datalength:	120 Bytes (0-120)
Return data description: Returns the FIRST datablock from an E-Ticket as a string	

Command Code	Command Flags
0x5C	0x40
Code (dec): 92	Flags (dec): 64

Get E-Ticket NEXT Block

16bit Identifier: 0x5C40

Gets the NEXT Datablock from already open E-Ticket
Note - Before running this command you must open (for reading) an E-Ticket in the memory of the TCS3000
Note - The length of all datablocks during E-Ticket transfer is the same (usually 120 symbols) except for the last one
Note - if you run this command right after opening an E-Ticket it will return the First datablock

Modes: ALL ✖ IDLE ✔ W&M ✔ DLV ✖ BUSY ✖

COMMAND: HOST -> TCS3000 (Downlink)

Datatype:	N/A
Datalength:	0 Bytes
Host data description: No data required from host	

ANSWER: TCS3000 -> HOST (Uplink)

Datatype:	String
Datalength:	120 Bytes (0-120)
Return data description: Returns the NEXT datablock from an E-Ticket as a string	

Command Code	Command Flags
0x5D	0x40
Code (dec): 93	Flags (dec): 64

Get E-Ticket Block NR

16bit Identifier: 0x5D40

Gets the Specified Datablock from already open E-Ticket
Note - Before running this command you must open (for reading) an E-Ticket in the memory of the TCS3000
Note - The length of all datablocks during E-Ticket transfer is the same (usually 120 symbols) except for the last one
Note - it is recommended to use this command for transferring whole tickets

Modes: ALL ✖ IDLE ✔ W&M ✔ DLV ✖ BUSY ✖

COMMAND: HOST -> TCS3000 (Downlink)

Datatype:	U32
Datalength:	4 Bytes
Host data description: E-Ticket datablock NR (0 based)	

ANSWER: TCS3000 -> HOST (Uplink)

Datatype:	String
Datalength:	120 Bytes (0-120)
Return data description: Returns the requested datablock from an E-Ticket as a string	

Command Code		Command Flags		RCLOSE E-Ticket	
0x5E		0x40		16bit Identifier: 0x5E40	
Code (dec): 94		Flags (dec): 64			
Closes the currently open E-Ticket and marks it Printed AND Transferred					
Note - Before running this command you must ROPEN and transfer an E-Ticket					
Note - This command requires that the host calculates a DOW CRC8 checksum over the whole data (but only the data without flags headers etc.) contained in the E-Ticket					
Modes: ALL ✖ IDLE ✔ W&M ✔ DLV ✖ BUSY ✖					
COMMAND: HOST -> TCS3000 (Downlink)					
Datatype:		U8			
Datalength:		1 Bytes			
Host data description: DOW CRC8 Calculated over the data of the E-Ticket					
ANSWER: TCS3000 -> HOST (Uplink)					
Datatype:		ACKNOWLEDGE			
Datalength:		0 Bytes			
Return data description: Command returns only ERRORCODE and STATUS					

Command Code		Command Flags		ROPEN LAST E-Ticket	
0x6B		0x40		16bit Identifier: 0x6B40	
Code (dec): 107		Flags (dec): 64			
Opens the LAST DELIVERY E-Ticket in the memory of TCS3000 and returns the number of datablocks in it					
Note - This command is similar to command 0x5A except that you don't have to specify ticket number					
Modes: ALL ✖ IDLE ✔ W&M ✔ DLV ✖ BUSY ✖					
COMMAND: HOST -> TCS3000 (Downlink)					
Datatype:		N/A			
Datalength:		0 Bytes			
Host data description: No data required from host					
ANSWER: TCS3000 -> HOST (Uplink)					
Datatype:		U32			
Datalength:		4 Bytes			
Return data description: Number of datablocks for transferring the ticket					

Command Code	Command Flags
0x6C	0x40
Code (dec): 108	Flags (dec): 64

Get TCS FW Build

16bit Identifier: 0x6C40

Gets the full Firmware Build number from the TCS3000

Modes: ALL ✓ IDLE ✓ W&M ✓ DLV ✓ BUSY ✓

COMMAND: HOST -> TCS3000 (Downlink)

Datatype:	N/A
Datalength:	0 Bytes
Host data description: No data required from host	

ANSWER: TCS3000 -> HOST (Uplink)

Datatype:	String
Datalength:	20 Bytes (0-20)
Return data description: The full Firmware Build number as a (AlphaNumeric) string	

Command Code	Command Flags
0x6D	0x40
Code (dec): 109	Flags (dec): 64

Get TCS FW Version

16bit Identifier: 0x6D40

Gets the Firmware Version number from the TCS3000

Modes: ALL ✓ IDLE ✓ W&M ✓ DLV ✓ BUSY ✓

COMMAND: HOST -> TCS3000 (Downlink)

Datatype:	N/A
Datalength:	0 Bytes
Host data description: No data required from host	

ANSWER: TCS3000 -> HOST (Uplink)

Datatype:	String
Datalength:	20 Bytes (0-20)
Return data description: The Firmware Version as a (AlphaNumeric) string	

CONFIDENTIAL

Command Code	Command Flags
0x6E	0x40
Code (dec): 110	Flags (dec): 64

Get TCS SW Build

16bit Identifier: 0x6E40

Gets the full Software Build number from the TCS3000

Modes:

ALL

IDLE

W&M

DLV

BUSY

COMMAND: HOST -> TCS3000 (Downlink)

Datatype:	N/A
Datalength:	0 Bytes
Host data description: No data required from host	

ANSWER: TCS3000 -> HOST (Uplink)

Datatype:	String
Datalength:	20 Bytes (0-20)
Return data description: The full Software Build number as a (AlphaNumeric) string	

Command Code	Command Flags
0x6F	0x40
Code (dec): 111	Flags (dec): 64

Get TCS SW Version

16bit Identifier: 0x6F40

Gets the Software Version number from the TCS3000

Modes:

ALL

IDLE

W&M

DLV

BUSY

COMMAND: HOST -> TCS3000 (Downlink)

Datatype:	N/A
Datalength:	0 Bytes
Host data description: No data required from host	

ANSWER: TCS3000 -> HOST (Uplink)

Datatype:	String
Datalength:	20 Bytes (0-20)
Return data description: The Software Version as a (AlphaNumeric) string	

CONFIDENTIAL

Command Code	Command Flags	
0x81	0x20	<u>Set Product Unit Price</u>
Code (dec): 129	Flags (dec): 32	16bit Identifier: 0x8120

Sets the value of the Unit Price for the current delivery

Note - You need to run this command AFTER a delivery has been already initialized otherwise the value will be overwritten during the initialization of the next delivery with the value stored in the system

Modes:

ALL ✖

IDLE ✔

W&M ✔

DLV ✖

BUSY ✖

COMMAND: HOST -> TCS3000 (Downlink)

Datatype:	Dbl
Datalength:	8 Bytes
Host data description: The value of the Unit Price (e.g. USD/GAL) as a Dbl	

ANSWER: TCS3000 -> HOST (Uplink)

Datatype:	ACKNOWLEDGE
Datalength:	0 Bytes
Return data description: Command returns only ERRORCODE and STATUS	

Command Code	Command Flags	
0x81	0x40	<u>Get Product Unit Price</u>
Code (dec): 129	Flags (dec): 64	16bit Identifier: 0x8140

Gets the current value of the Unit Price

Note - If you run this command before a delivery has been initialized the value used during the delivery might be different from the one this command has returned

Modes:

ALL ✖

IDLE ✔

W&M ✔

DLV ✖

BUSY ✖

COMMAND: HOST -> TCS3000 (Downlink)

Datatype:	N/A
Datalength:	0 Bytes
Host data description: No data required from host	

ANSWER: TCS3000 -> HOST (Uplink)

Datatype:	Dbl
Datalength:	8 Bytes
Return data description: Returns the value of Unit Price as a Dbl	

Command Code	Command Flags	
0x82	0x20	<u>Set Product Tax 1 Value</u>
Code (dec): 130	Flags (dec): 32	16bit Identifier: 0x8220

Sets the value of Tax 1 for the current delivery

Note - You need to run this command AFTER a delivery has been already initialized otherwise the value will be overwritten during the initialization of the next delivery with the value stored in the system

Modes: ALL ✖ IDLE ✔ W&M ✔ DLV ✖ BUSY ✖

COMMAND: HOST -> TCS3000 (Downlink)

Datatype:	Dbl
Datalength:	8 Bytes
Host data description: Value of Tax 1 as Dbl (in case the tax unit is % use 0.xx e.g. 0.15=15%)	

ANSWER: TCS3000 -> HOST (Uplink)

Datatype:	ACKNOWLEDGE
Datalength:	0 Bytes
Return data description: Command returns only ERRORCODE and STATUS	

Command Code	Command Flags	
0x82	0x40	<u>Get Product Tax 1 Value</u>
Code (dec): 130	Flags (dec): 64	16bit Identifier: 0x8240

Gets the current value of Tax 1

Note - If you run this command before a delivery has been initialized the value used during the delivery might be different from the one this command has returned

Modes: ALL ✖ IDLE ✔ W&M ✔ DLV ✖ BUSY ✖

COMMAND: HOST -> TCS3000 (Downlink)

Datatype:	N/A
Datalength:	0 Bytes
Host data description: No data required from host	

ANSWER: TCS3000 -> HOST (Uplink)

Datatype:	Dbl
Datalength:	8 Bytes
Return data description: Value of Tax 1 as Dbl (in case the tax unit is % returns 0.xx e.g. 0.15=15%)	

CONFIDENTIAL

Command Code	Command Flags	
0x83	0x20	<u>Set Product Tax 2 Value</u>
Code (dec): 131	Flags (dec): 32	16bit Identifier: 0x8320

Sets the value of Tax 2 for the current delivery

Note - You need to run this command AFTER a delivery has been already initialized otherwise the value will be overwritten during the initialization of the next delivery with the value stored in the system

Modes: ALL ✖ IDLE ✔ W&M ✔ DLV ✖ BUSY ✖

COMMAND: HOST -> TCS3000 (Downlink)

Datatype:	Dbl
Datalength:	8 Bytes
Host data description: Value of Tax 2 as Dbl (in case the tax unit is % use 0.xx e.g. 0.15=15%)	

ANSWER: TCS3000 -> HOST (Uplink)

Datatype:	ACKNOWLEDGE
Datalength:	0 Bytes
Return data description: Command returns only ERRORCODE and STATUS	

Command Code	Command Flags	
0x83	0x40	<u>Get Product Tax 2 Value</u>
Code (dec): 131	Flags (dec): 64	16bit Identifier: 0x8340

Gets the current value of Tax 2

Note - If you run this command before a delivery has been initialized the value used during the delivery might be different from the one this command has returned

Modes: ALL ✖ IDLE ✔ W&M ✔ DLV ✖ BUSY ✖

COMMAND: HOST -> TCS3000 (Downlink)

Datatype:	N/A
Datalength:	0 Bytes
Host data description: No data required from host	

ANSWER: TCS3000 -> HOST (Uplink)

Datatype:	Dbl
Datalength:	8 Bytes
Return data description: Value of Tax 2 as Dbl (in case the tax unit is % returns 0.xx e.g. 0.15=15%)	

Command Code	Command Flags	
0x84	0x20	<u>Set Product Tax 1 Text</u>
Code (dec): 132	Flags (dec): 32	16bit Identifier: 0x8420

Sets the Text Name of Tax 1 for the current delivery
Note - You need to run this command AFTER a delivery has been already initialized otherwise the value will be overwritten during the initialization of the next delivery with the value stored in the system

Modes: ALL ✖ IDLE ✔ W&M ✔ DLV ✖ BUSY ✖

COMMAND: HOST -> TCS3000 (Downlink)

Datatype:	String	
Datalength:	20 Bytes	(0-20)
Host data description: Name of Tax 1 as String		

ANSWER: TCS3000 -> HOST (Uplink)

Datatype:	ACKNOWLEDGE	
Datalength:	0 Bytes	
Return data description: Command returns only ERRORCODE and STATUS		

Command Code	Command Flags	
0x84	0x40	<u>Get Product Tax 1 Text</u>
Code (dec): 132	Flags (dec): 64	16bit Identifier: 0x8440

Gets the current Text Name of Tax 1
Note - If you run this command before a delivery has been initialized the value used during the delivery might be different from the one this command has returned

Modes: ALL ✖ IDLE ✔ W&M ✔ DLV ✖ BUSY ✖

COMMAND: HOST -> TCS3000 (Downlink)

Datatype:	N/A	
Datalength:	0 Bytes	
Host data description: No data required from host		

ANSWER: TCS3000 -> HOST (Uplink)

Datatype:	String	
Datalength:	20 Bytes	(0-20)
Return data description: Name of Tax 1 as String		

Command Code	Command Flags	
0x85	0x20	<u>Set Product Tax 2 Text</u>
Code (dec): 133	Flags (dec): 32	16bit Identifier: 0x8520

Sets the Text Name of Tax 2 for the current delivery
Note - You need to run this command AFTER a delivery has been already initialized otherwise the value will be overwritten during the initialization of the next delivery with the value stored in the system

Modes: ALL ✖ IDLE ✔ W&M ✔ DLV ✖ BUSY ✖

COMMAND: HOST -> TCS3000 (Downlink)

Datatype:	String	
Datalength:	20 Bytes	(0-20)
Host data description: Name of Tax 2 as String		

ANSWER: TCS3000 -> HOST (Uplink)

Datatype:	ACKNOWLEDGE	
Datalength:	0 Bytes	
Return data description: Command returns only ERRORCODE and STATUS		

Command Code	Command Flags	
0x85	0x40	<u>Get Product Tax 2 Text</u>
Code (dec): 133	Flags (dec): 64	16bit Identifier: 0x8540

Gets the current Text Name of Tax 2
Note - If you run this command before a delivery has been initialized the value used during the delivery might be different from the one this command has returned

Modes: ALL ✖ IDLE ✔ W&M ✔ DLV ✖ BUSY ✖

COMMAND: HOST -> TCS3000 (Downlink)

Datatype:	N/A	
Datalength:	0 Bytes	
Host data description: No data required from host		

ANSWER: TCS3000 -> HOST (Uplink)

Datatype:	String	
Datalength:	20 Bytes	(0-20)
Return data description: Name of Tax 2 as String		

Command Code	Command Flags	
0x86	0x20	<u>Set Product Tax 1 Type</u>
Code (dec): 134	Flags (dec): 32	16bit Identifier: 0x8620

Sets the Type of Tax 1 for the current delivery

Note - You need to run this command AFTER a delivery has been already initialized otherwise the value will be overwritten during the initialization of the next delivery with the value stored in the system

Modes:

ALL ✖

IDLE ✔

W&M ✔

DLV ✖

BUSY ✖

COMMAND: HOST -> TCS3000 (Downlink)

Datatype:	ENUM
Datalength:	1 Bytes
Host data description: ENUM (TAX1_TYPE) 0=ERROR 1=NONE 2=PERCENT 3=PERUNIT 4=TAXTAX (Tax on Tax - % calculated over the total value)	

ANSWER: TCS3000 -> HOST (Uplink)

Datatype:	ACKNOWLEDGE
Datalength:	0 Bytes
Return data description: Command returns only ERRORCODE and STATUS	

Command Code	Command Flags	
0x86	0x40	<u>Get Product Tax 1 Type</u>
Code (dec): 134	Flags (dec): 64	16bit Identifier: 0x8640

Gets the current Type of Tax 1

Note - If you run this command before a delivery has been initialized the value used during the delivery might be different from the one this command has returned

Modes:

ALL ✖

IDLE ✔

W&M ✔

DLV ✖

BUSY ✖

COMMAND: HOST -> TCS3000 (Downlink)

Datatype:	N/A
Datalength:	0 Bytes
Host data description: No data required from host	

ANSWER: TCS3000 -> HOST (Uplink)

Datatype:	ENUM
Datalength:	1 Bytes
Return data description: ENUM (TAX1_TYPE) 0=ERROR 1=NONE 2=PERCENT 3=PERUNIT 4=TAXTAX (Tax on Tax - % calculated over the total value)	

Command Code

0x87

Code (dec): 135

Command Flags

0x20

Flags (dec): 32

Set Product Tax 2 Type

16bit Identifier: 0x8720

Sets the Type of Tax 2 for the current delivery

Note - You need to run this command AFTER a delivery has been already initialized

otherwise the value will be overwritten during the initialization of the next delivery with the value stored in the system

Modes: ALL ☐ IDLE ☒ W&M ☒ DLV ☐ BUSY ☐

COMMAND: HOST -> TCS3000 (Downlink)

Datatype:	ENUM
Datalength:	1 Bytes
Host data description: ENUM (TAX2_TYPE)	
0=ERROR	
1=NONE	
2=PERCENT	
3=PERUNIT	
4=TAXTAX (Tax on Tax - % calculated over the total value incl. Tax 1)	

ANSWER: TCS3000 -> HOST (Uplink)

Datatype:	ACKNOWLEDGE
Datalength:	0 Bytes
Return data description: Command returns only ERRORCODE and STATUS	

Command Code

0x87

Code (dec): 135

Command Flags

0x40

Flags (dec): 64

Get Product Tax 2 Type

16bit Identifier: 0x8740

Gets the current Type of Tax 2

Note - If you run this command before a delivery has been initialized the value used during the delivery

might be different from the one this command has returned

Modes: ALL ☐ IDLE ☒ W&M ☒ DLV ☐ BUSY ☐

COMMAND: HOST -> TCS3000 (Downlink)

Datatype:	N/A
Datalength:	0 Bytes
Host data description: No data required from host	

ANSWER: TCS3000 -> HOST (Uplink)

Datatype:	ENUM
Datalength:	1 Bytes
Return data description: ENUM (TAX2_TYPE)	
0=ERROR	
1=NONE	
2=PERCENT	
3=PERUNIT	
4=TAXTAX (Tax on Tax - % calculated over the total value incl. Tax 1)	

Command Code	Command Flags	
0x90	0x20	Load Products Table
Code (dec): 144	Flags (dec): 32	16bit Identifier: 0x9020

Re-Loads the products table in the memory of TCS3000
*Note - the product table is automatically loaded during TCS3000 Startup
you need this command only if there are changes to the products.*

Modes: ALL ☐ IDLE ☒ W&M ☒ DLV ☐ BUSY ☐

COMMAND: HOST -> TCS3000 (Downlink)

Datatype:	N/A
Datalength:	0 Bytes
Host data description: No data required from host	

ANSWER: TCS3000 -> HOST (Uplink)

Datatype:	U16
Datalength:	2 Bytes
Return data description: Number of products in the product table	

Command Code	Command Flags	
0x91	0x40	Get Number of Products
Code (dec): 145	Flags (dec): 64	16bit Identifier: 0x9140

Returns the number of products in the product table of TCS3000

Modes: ALL ☒ IDLE ☒ W&M ☒ DLV ☒ BUSY ☒

COMMAND: HOST -> TCS3000 (Downlink)

Datatype:	N/A
Datalength:	0 Bytes
Host data description: No data required from host	

ANSWER: TCS3000 -> HOST (Uplink)

Datatype:	U16
Datalength:	2 Bytes
Return data description: Number of products in the product table	

Command Code

0x92

Code (dec): 146

Command Flags

0x40

Flags (dec): 64

Get First Product ID

16bit Identifier: 0x9240

Returns the FIRST Product ID (PRODID U16) and Product Status from the product table of TCS3000

Modes: ALL ✓ IDLE ✓ W&M ✓ DLV ✓ BUSY ✓

COMMAND: HOST -> TCS3000 (Downlink)

Datatype:

N/A

Datalength:

0 Bytes

Host data description: No data required from host

ANSWER: TCS3000 -> HOST (Uplink)

Datatype:

String

Datalength:

3 Bytes

Return data description: First Product ID and Product status as Combined numeric data:
<ProdID><ProdStatus>
<U16><U8>
ProdStatus:
0=Error/Not Calibrated
1=Calibrated Not Active
3=Calibrated and Active (The only type of product that can be used for delivery)

Command Code

0x93

Code (dec): 147

Command Flags

0x40

Flags (dec): 64

Get Next Product ID

16bit Identifier: 0x9340

Returns the NEXT Product ID (PRODID U16) and Product Status from the product table of TCS3000

Note - After startup or command 0x90 this command returns the First Product ID

Modes: ALL ✓ IDLE ✓ W&M ✓ DLV ✓ BUSY ✓

COMMAND: HOST -> TCS3000 (Downlink)

Datatype:

N/A

Datalength:

0 Bytes

Host data description: No data required from host

ANSWER: TCS3000 -> HOST (Uplink)

Datatype:

String

Datalength:

3 Bytes

Return data description: Next Product ID and Product status as Combined numeric data:
<ProdID><ProdStatus>
<U16><U8>
ProdStatus:
0=Error/Not Calibrated
1=Calibrated Not Active
3=Calibrated and Active (The only type of product that can be used for delivery)

Command Code	Command Flags	
0x94	0x40	Get Product ID
Code (dec): 148	Flags (dec): 64	16bit Identifier: 0x9440

Returns the specified Product ID (PRODID U16) and Product Status from the product table of TCS3000

Note - it is recommended to use this command for transferring the whole product table

Modes: ALL ✓ IDLE ✓ W&M ✓ DLV ✓ BUSY ✓

COMMAND: HOST -> TCS3000 (Downlink)

Datatype:	U16
Datalength:	2 Bytes
Host data description: Index of the product in the product table (0 based)	

ANSWER: TCS3000 -> HOST (Uplink)

Datatype:	String
Datalength:	3 Bytes
Return data description: Product ID and Product status as Combined numeric data: <ProdID><ProdStatus> <U16><U8> ProdStatus: 0=Error/Not Calibrated 1=Calibrated Not Active 3=Calibrated and Active (The only type of product that can be used for delivery)	

CONFIDENTIAL

Command Code	Command Flags
0x95	0x40
Code (dec): 149	Flags (dec): 64

Get First Product Name

16bit Identifier: 0x9540

Returns the FIRST Product name from the product table of TCS3000

Modes: ALL ✓ IDLE ✓ W&M ✓ DLV ✓ BUSY ✓

COMMAND: HOST -> TCS3000 (Downlink)

Datatype:	N/A
Datalength:	0 Bytes
Host data description: No data required from host	

ANSWER: TCS3000 -> HOST (Uplink)

Datatype:	String
Datalength:	20 Bytes (0-20)
Return data description: The First Product name as String	

Command Code	Command Flags
0x96	0x40
Code (dec): 150	Flags (dec): 64

Get Next Product Name

16bit Identifier: 0x9640

Returns the Next Product name from the product table of TCS3000

Note - After startup or command 0x90 this command returns the First Product Name

Modes: ALL ✓ IDLE ✓ W&M ✓ DLV ✓ BUSY ✓

COMMAND: HOST -> TCS3000 (Downlink)

Datatype:	N/A
Datalength:	0 Bytes
Host data description: No data required from host	

ANSWER: TCS3000 -> HOST (Uplink)

Datatype:	String
Datalength:	20 Bytes (0-20)
Return data description: The Next Product name as String	

Command Code	Command Flags
0x97	0x40
Code (dec): 151	Flags (dec): 64

Get Product Name

16bit Identifier: 0x9740

Returns the selected Product name from the product table of TCS3000

Note - it is recommended to use this command when fetching all product names from the product table

Modes: ALL ✓ IDLE ✓ W&M ✓ DLV ✓ BUSY ✓

COMMAND: HOST -> TCS3000 (Downlink)

Datatype:	U16
Datalength:	2 Bytes
Host data description: Index of the product in the product table (0 based)	

ANSWER: TCS3000 -> HOST (Uplink)

Datatype:	String
Datalength:	20 Bytes (0-20)
Return data description: The First Product name as String	

Command Code

0x98

Code (dec): 152

Command Flags

0x40

Flags (dec): 64

Get Product Name by ID

16bit Identifier: 0x9840

Returns the selected (as PRODID) Product name from the product table of TCS3000

Note - it is recommended to use this command when fetching all product names from the product table

Modes: ALL ✓ IDLE ✓ W&M ✓ DLV ✓ BUSY ✓

COMMAND: HOST -> TCS3000 (Downlink)

Datatype:	String	
Datalength:	120 Bytes	(0-120)
Host data description: Product ID (U16) Range: 1001-9999		

ANSWER: TCS3000 -> HOST (Uplink)

Datatype:	String	
Datalength:	20 Bytes	(0-20)
Return data description: The Product name as String		

CONFIDENTIAL

Command Code

0x9E

Code (dec): 158

Command Flags

0x20

Flags (dec): 32

Write G-Ticket First Line

16bit Identifier: 0x9E20

Writes the FIRST Line of the G-Ticket in the memory of TCS3000

Note - NOT RECOMMENDED FOR NORMAL USE! Use command 0xA1 instead

Modes: ALL ✗ IDLE ✓ W&M ✓ DLV ✗ BUSY ✗

COMMAND: HOST -> TCS3000 (Downlink)

Datatype: String

Datalength: 120 Bytes (0-120)

Host data description: The FIRST Line of the G-Ticket as string

ANSWER: TCS3000 -> HOST (Uplink)

Datatype: ACKNOWLEDGE

Datalength: 0 Bytes

Return data description: Command returns only ERRORCODE and STATUS

Command Code

0x9F

Code (dec): 159

Command Flags

0x20

Flags (dec): 32

Write G-Ticket Next Line

16bit Identifier: 0x9F20

Writes the Next Line of the G-Ticket in the memory of TCS3000

Note - if executed right after command 0xA0 (WOPEN G-Ticket) this command writes the first line

Note - NOT RECOMMENDED FOR NORMAL USE! Use command 0xA1 instead

Modes: ALL ✗ IDLE ✓ W&M ✓ DLV ✗ BUSY ✗

COMMAND: HOST -> TCS3000 (Downlink)

Datatype: String

Datalength: 120 Bytes (0-120)

Host data description: The Next Line of the G-Ticket as string

ANSWER: TCS3000 -> HOST (Uplink)

Datatype: ACKNOWLEDGE

Datalength: 0 Bytes

Return data description: Command returns only ERRORCODE and STATUS

CONFIDENTIAL

Command Code	Command Flags
0xA0	0x20
Code (dec): 160	Flags (dec): 32

WOPEN G-Ticket

16bit Identifier: 0xA020

Initializes empty G-Ticket in the memory of TCS3000 with the specified size (lines and columns)

Modes: ALL ✖ IDLE ✔ W&M ✔ DLV ✖ BUSY ✖

COMMAND: HOST -> TCS3000 (Downlink)

Datatype:	String
Datalength:	4 Bytes
Host data description: The size of the G-Ticket as string - Combined numeric data: <NrLines><LineLength> <U16><U16>	

ANSWER: TCS3000 -> HOST (Uplink)

Datatype:	ACKNOWLEDGE
Datalength:	0 Bytes
Return data description: Command returns only ERRORCODE and STATUS	

Command Code	Command Flags
0xA1	0x20
Code (dec): 161	Flags (dec): 32

WRITE G-Ticket

16bit Identifier: 0xA120

Writes the specified G-Ticket line in the memory of TCS3000

Modes: ALL ✖ IDLE ✔ W&M ✔ DLV ✖ BUSY ✖

COMMAND: HOST -> TCS3000 (Downlink)

Datatype:	String
Datalength:	120 Bytes (1-120)
Host data description: The Line number (0 based) and line content as string - Combined data: <LineNR><LineText> <U16><String>	

ANSWER: TCS3000 -> HOST (Uplink)

Datatype:	ACKNOWLEDGE
Datalength:	0 Bytes
Return data description: Command returns only ERRORCODE and STATUS	

Command Code	Command Flags
0xA2	0x20
Code (dec): 162	Flags (dec): 32

WCLOSE G-Ticket

16bit Identifier: 0xA220

Closes the G-Ticket and stores it in the TCS3000 (it can be printed after that)

Modes: ALL ✖ IDLE ✔ W&M ✔ DLV ✖ BUSY ✖

COMMAND: HOST -> TCS3000 (Downlink)

Datatype:	U8
Datalength:	1 Bytes
Host data description: The DOW CRC8 calculated over the whole data of the G-Ticket	

ANSWER: TCS3000 -> HOST (Uplink)

Datatype:	ACKNOWLEDGE
Datalength:	0 Bytes
Return data description: Command returns only ERRORCODE and STATUS	

Command Code	Command Flags
0xA4	0x40
Code (dec): 164	Flags (dec): 64

ROPEN G-Ticket

16bit Identifier: 0xA440

Loads the currently stored G-Ticket to the memory of TCS3000 and returns the number of lines in it

Modes: ALL ✖ IDLE ✔ W&M ✖ DLV ✖ BUSY ✖

COMMAND: HOST -> TCS3000 (Downlink)

Datatype:	N/A
Datalength:	0 Bytes
Host data description: No data required from host	

ANSWER: TCS3000 -> HOST (Uplink)

Datatype:	U16
Datalength:	2 Bytes
Return data description: Number of lines in the G-Ticket	

Command Code	Command Flags
0xA5	0x40
Code (dec): 165	Flags (dec): 64

Get G-Ticket Next Line

16bit Identifier: 0xA540

Gets the Next line from the G-Ticket loaded in the memory (by command 0xA4) of TCS3000
Note - if executed right after command 0xA4 (ROPEN G-Ticket) this command gets the first line

Modes: ALL ✖ IDLE ✔ W&M ✖ DLV ✖ BUSY ✖

COMMAND: HOST -> TCS3000 (Downlink)

Datatype:	N/A
Datalength:	0 Bytes
Host data description: No data required from host	

ANSWER: TCS3000 -> HOST (Uplink)

Datatype:	U16
Datalength:	120 Bytes (0-120)
Return data description: The content of the Next G-Ticket line as a string	

Command Code	Command Flags
0xA6	0x40
Code (dec): 166	Flags (dec): 64

Get G-Ticket Line NR

16bit Identifier: 0xA640

Gets the requested line from the G-Ticket loaded in the memory (by command 0xA4) of TCS3000
Note - this is the recommended command when getting the whole G-Ticket

Modes: ALL ✖ IDLE ✔ W&M ✖ DLV ✖ BUSY ✖

COMMAND: HOST -> TCS3000 (Downlink)

Datatype:	U16
Datalength:	2 Bytes
Host data description: Line number as U16	

ANSWER: TCS3000 -> HOST (Uplink)

Datatype:	U16
Datalength:	120 Bytes (0-120)
Return data description: The content of the requested G-Ticket line as a string	

Command Code

0xA7

Code (dec): 167

Command Flags

0x40

Flags (dec): 64

Get G-Ticket First Line

16bit Identifier: 0xA740

Gets the First line from the G-Ticket loaded in the memory (by command 0xA4) of TCS3000

Modes:

ALL ✖

IDLE ✔

W&M ✖

DLV ✖

BUSY ✖

COMMAND: HOST -> TCS3000 (Downlink)

Datatype:

N/A

Datalength:

0 Bytes

Host data description: No data required from host

ANSWER: TCS3000 -> HOST (Uplink)

Datatype:

U16

Datalength:

120 Bytes

(0-120)

Return data description: The content of the First G-Ticket line as a string

CONFIDENTIAL

Command Code	Command Flags
0xA8	0x20
Code (dec): 168	Flags (dec): 32

Print G-Ticket

16bit Identifier: 0xA820

Prints the G-Ticket stored inside the TCS3000
Note - this command will send an acknowledgement immediately - it will not wait for the printing process to finish!

Modes: ALL ✖ IDLE ✔ W&M ✖ DLV ✖ BUSY ✖

COMMAND: HOST -> TCS3000 (Downlink)

Datatype:	N/A
Datalength:	0 Bytes
Host data description: No data required from host	

ANSWER: TCS3000 -> HOST (Uplink)

Datatype:	ACKNOWLEDGE
Datalength:	0 Bytes
Return data description: Command returns only ERRORCODE and STATUS	

Command Code	Command Flags
0xA9	0x20
Code (dec): 169	Flags (dec): 32

Clear Last Ticket Status

16bit Identifier: 0xA920

Clears the "Not Printed" status of the ticket from the last delivery and changes it's type to PRINTED
Note - if the last ticket is already printed (or has printed status) or Transferred this command does not modify it.

Modes: ALL ✖ IDLE ✔ W&M ✖ DLV ✖ BUSY ✖

COMMAND: HOST -> TCS3000 (Downlink)

Datatype:	N/A
Datalength:	0 Bytes
Host data description: No data required from host	

ANSWER: TCS3000 -> HOST (Uplink)

Datatype:	ACKNOWLEDGE
Datalength:	0 Bytes
Return data description: Command returns only ERRORCODE and STATUS	

Command Code	Command Flags	
0xF0	0x20	Force IDLE Mode
Code (dec): 240	Flags (dec): 32	16bit Identifier: 0xF020

Forces the system to go to IDLE (show last delivery) mode
Use carefully because this command will force the system to go to IDLE mode disregarding any unsaved data!
This is not a very "gentle" command and should be used with care
This command WILL NOT Execute during active delivery

Modes:

ALL ✖

IDLE ✔

W&M ✔

DLV ✖

BUSY ✔

COMMAND: HOST -> TCS3000 (Downlink)

Datatype:	ENUM
Datalength:	1 Bytes
Host data description: ENUM (RI_IDLETYPE) 0=ERROR 1=IDLE 2=WMCONFIG (Remote W&M Mode)	

ANSWER: TCS3000 -> HOST (Uplink)

Datatype:	ACKNOWLEDGE
Datalength:	0 Bytes
Return data description: Command returns only ERRORCODE and STATUS	

CONFIDENTIAL

Command Code	Command Flags	
0xF6	0x20	System SHUTDOWN
Code (dec): 246	Flags (dec): 32	16bit Identifier: 0xF620

Shuts down the TCS3000 system and powers it off

Note: This command needs to be executed in two steps:

- 1) Run this command with parameter value 1 (INITIATE)
- 2) Within 10s run this command again with parameter value 2 or 3 depending on what you need

Modes: ALL ✖ IDLE ✔ W&M ✖ DLV ✖ BUSY ✖

COMMAND: HOST -> TCS3000 (Downlink)

Datatype:	ENUM
Datalength:	1 Bytes
Host data description: ENUM (SYS_SHUTDOWN) 0=ERROR 1=INITIATE 2=CONFIRM_NOSAVE (Shuts down WITHOUT saving the system state) 3=CONFIRM_SAVESTATE (Shuts down after saving the system state first)	

ANSWER: TCS3000 -> HOST (Uplink)

Datatype:	ACKNOWLEDGE
Datalength:	0 Bytes
Return data description: Command returns only ERRORCODE and STATUS	

Command Code	Command Flags	
0xF7	0x20	System REBOOT
Code (dec): 247	Flags (dec): 32	16bit Identifier: 0xF720

Restarts the TCS3000 system

Note: This command needs to be executed in two steps:

- 1) Run this command with parameter value 1 (INITIATE)
- 2) Within 10s run this command again with parameter value 2 or 3 depending on what you need

Modes: ALL ✖ IDLE ✔ W&M ✖ DLV ✖ BUSY ✖

COMMAND: HOST -> TCS3000 (Downlink)

Datatype:	ENUM
Datalength:	1 Bytes
Host data description: ENUM (SYS_SHUTDOWN) 0=ERROR 1=INITIATE 2=CONFIRM_NOSAVE (Reboots WITHOUT saving the system state) 3=CONFIRM_SAVESTATE (Reboots after saving the system state first)	

ANSWER: TCS3000 -> HOST (Uplink)

Datatype:	ACKNOWLEDGE
Datalength:	0 Bytes
Return data description: Command returns only ERRORCODE and STATUS	

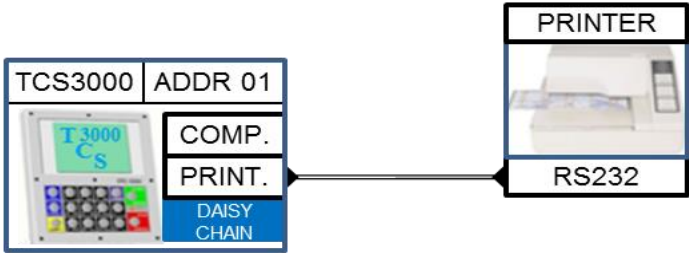
TCS3000 REMOTE INTERFACE ERROR CODES

ERRCODE	(HEX)	ERROR CODE DESCRIPTION
0	0x00	NO ERROR
1	0x01	COMMAND NOT RECOGNIZED
2	0x02	DS READ - DBL VALUE is NaN
3	0x03	DS READ - RETURNED INVALID NUMERIC DATA VALUE
4	0x04	DS READ - ENUM VALUE NOT RECOGNIZED
5	0x05	DS READ - VARIABLE NOT FOUND OR READ ERROR
6	0x06	DS WRITE - ERROR DURING INTERNAL DATA WRITE OPERATION
7	0x07	INCORRECT COMMAND FORMAT
8	0x08	REPEAT COMMAND REQUESTED AS FIRST COMMAND!
10	0x0A	RESERVED
11	0x0B	SYSTEM RETURNED INVALID DELIVERY STATE
15	0x0F	TIMER NOT DEFINED OR ZERO TIMEOUT TARGET
19	0x13	INCORRECT PASSWORD
20	0x14	SECONDARY ENUM VALUE OUT OF RANGE
21	0x15	INCORRECT BOOLEAN VALUE FORMAT
27	0x1B	RESERVED
31	0x1F	ERROR DURING PRODUCT TABLE LOADING
32	0x20	PRODUCT TABLE NOT LOADED IN MEMORY
33	0x21	PRODUCT TABLE INDEX OUT OF BOUNDS
34	0x22	PRODUCT ID NOT FOUND IN THE PRODUCT TABLE
35	0x23	NO ACTIVE SHIFT
36	0x24	LAST DELIVERY TICKET STATUS NOT OK
37	0x25	NON-PRINTED TICKETS MEMORY FULL! ABORTING DELIVERY INITIALIZATION!
48	0x30	ERROR DURING DIRECT DELIVERY INIT
49	0x31	ERROR DURING PRESET DELIVERY INIT
50	0x32	DELIVERY NOT CONFIGURED
51	0x33	START DELIVERY TYPE DOES NOT MATCH THE CONFIGURED DELIVERY TYPE
59	0x3B	PRESET DELIVERY IS NOT ALLOWED
60	0x3C	REQUESTED NET PRESET ON NONCOMPENSATED PRODUCT
61	0x3D	PRICE PRESET NOT ALLOWED ON ZERO PRICE PRODUCT
62	0x3E	REQUESTED PRESET AMOUNT OR VOLUME IS ZERO, NEGATIVE OR TOO SMALL
63	0x3F	INCORRECT INPUT NUMERIC VALUE
64	0x40	INCORRECT INPUT DATA LENGTH
65	0x41	ENUM VALUE OUT OF RANGE
66	0x42	PRODUCT_ID OUT OF RANGE (1000-9999)
67	0x43	PRODUCT_ID NOT FOUND OR NOT ACTIVE
68	0x44	NO ACTIVE PRODUCTS
69	0x45	ERROR DURING PRODUCT DATA LOADING
70	0x46	ERROR DURING PRODUCTS CONFIGURATION READING
71	0x47	TOTALIZER VALUE FOR THIS PRODUCT ID CANNOT BE FOUND
72	0x48	INVALID TIME VALUE
73	0x49	INVALID DATE VALUE
80	0x50	ETICKETS TABLE EMPTY OR NO INITIALIZED
81	0x51	ETICKETS TABLE INDEX OUT OF BOUNDS
82	0x52	ETICKET NOT FOUND
83	0x53	ERROR WHILE LOADING ETICKET
84	0x54	ETICKET DATA NOT LOADED
85	0x55	ETICKET BLOCKNR OUT OF BOUNDS
86	0x56	ETICKET DATA CRC ERROR
87	0x57	INVALID STICKET DEFINITION REQUESTED OR DEFINITION CORRUPTED
88	0x58	STICKET DATA MISSING OR INVALID
96	0x60	OPERATION CANNOT BE EXECUTED IN CURRENT SYSTEM MODE
97	0x61	CALIBRATION BOLT NOT IN PLACE
98	0x62	CALIBRATION BOLT IN PLACE
101	0x65	TEMPERATURE SENSOR READING ERROR
102	0x66	TEMPERATURE SENSOR CALIBRATION FAILED
126	0x7E	RESERVED
128	0x80	INVALID COMMAND DEFINITION
129	0x81	DS ERROR - ERROR DURING DS READING
130	0x82	DS READ - DOUBLE RETURNED NON NUMERIC VALUE
131	0x83	ENUM DEFINITION MISSING FROM TCS3000 RI CONFIGURATION
132	0x84	ERROR IN ENUM DEFINITION IN TCS3000
140	0x8C	USR_DB INDEX OUT OF BOUNDS
141	0x8D	USR_DB ACCESS ERROR
142	0x8E	USR_DB RECORD NOT FOUND
143	0x8F	USR_DB COMMAND FORMATTED INCORRECTLY
144	0x90	USR_DB TOO MANY MODIFIED RECORDS - FULL DB DOWNLOAD REQUIRED!
145	0x91	USR_DB TOO MANY MODIFIED RECORDS - RECORD CHANGE NOT REGISTERED!
148	0x94	CUSTID IS REQUIRED AND CANNOT BE EMPTY
149	0x95	CANNOT START DELIVERY - ONE OR MORE REQUIRED CUSTID IS NOT SET CORRECTLY
150	0x96	REQUESTED CUST ID NOT FOUND
151	0x97	REQUESTED CUST ID FOUND BUT NOT VALID (EXPIRED OR NOT ACTIVE)
160	0xA0	DS ERROR - ERROR DURING DS WRITING
161	0xA1	COMMAND ERROR - THE SYSTEM RETURNED ERROR DURING COMMAND EXECUTION
162	0xA2	SYSTEM ERROR - TIMEOUT DURING COMMAND EXECUTION
163	0xA3	SYSTEM ERROR - INCORRECT INTERNAL COMMAND ACKNOWLEDGMENT
180	0xB4	GTKT LINE INDEX OUT OF BOUNDS
181	0xB5	GTKT NOT IN WRITE MODE
182	0xB6	INCORRECT GTKT LINE LENGTH
183	0xB7	ERROR WHILE SETTING GTKT LINE
185	0xB9	ZERO DIMENTION IN GTKT DATA
186	0xBA	GTKT CHECKSUM ERROR
187	0xBB	ERROR WHILE SAVING GTKT
245	0xF5	CURRENT PASSWORD NOT ENTERED OR 10s TIMEOUT EXPIRED
246	0xF6	SHUTDOWN NOT INITIATED OR 10s TIMEOUT EXPIRED
247	0xF7	REBOOT NOT INITIATED OR 10s TIMEOUT EXPIRED
250	0xFA	EXTENDED ERRORCODE (SEE STATUS BYTE FOR CODE)
252	0xFC	INTEFACE BUSY
253	0xFD	SYSTEM BUSY
254	0xFE	SYSTEM SHUTDOWN IN PROGRESS
255	0xFF	DATA INTEGRITY ERROR

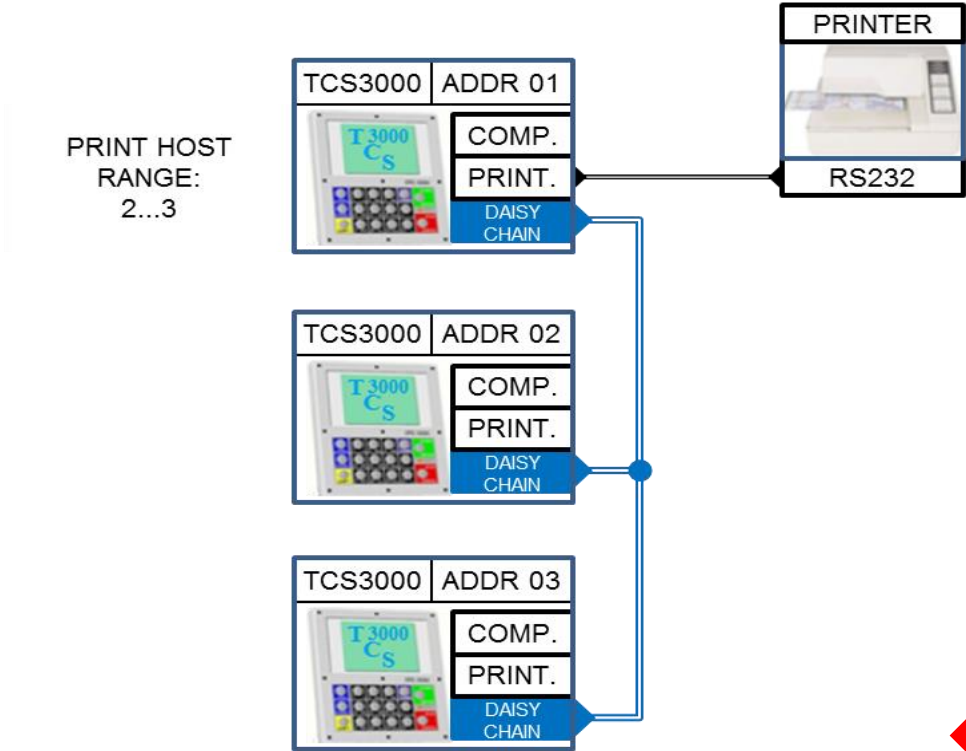
TCS3000 REMOTE INTERFACE CONNECTION EXAMPLES

CONFIDENTIAL

EXAMPLE CONNECTION 1: STANDALONE DEVICE

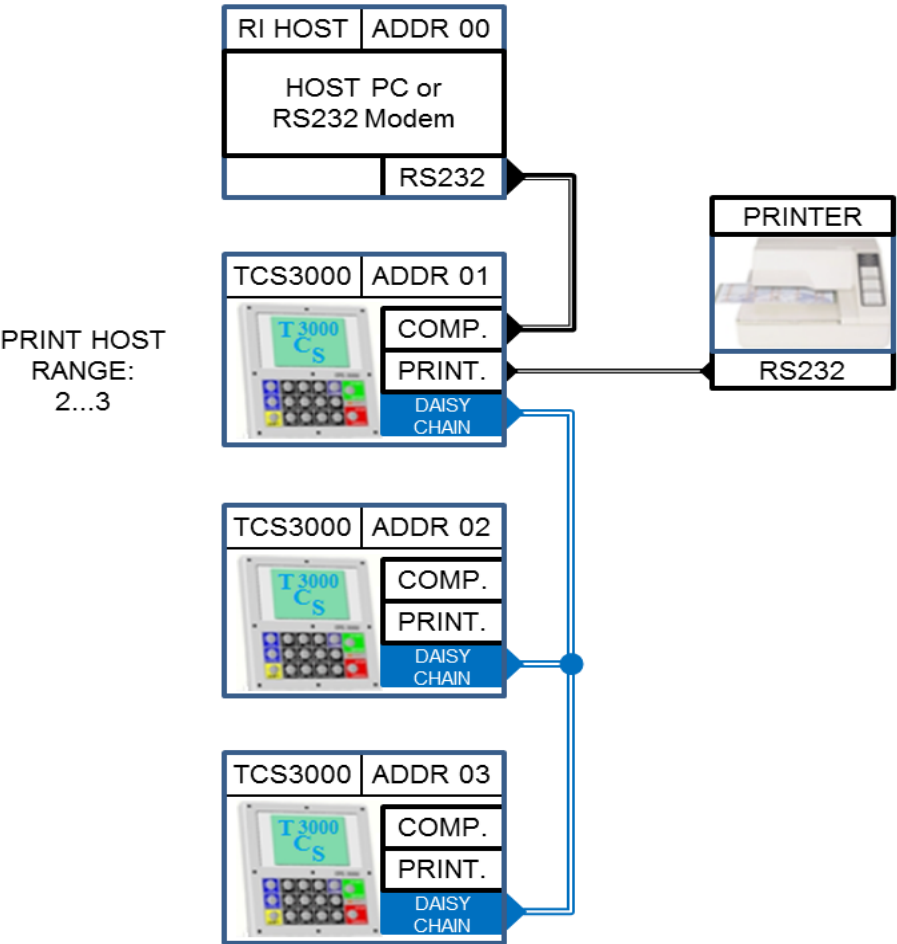


EXAMPLE CONNECTION 2: STANDALONE 3 DEVICES PRINT CHAIN WITH TCS3000 PRINT HOST



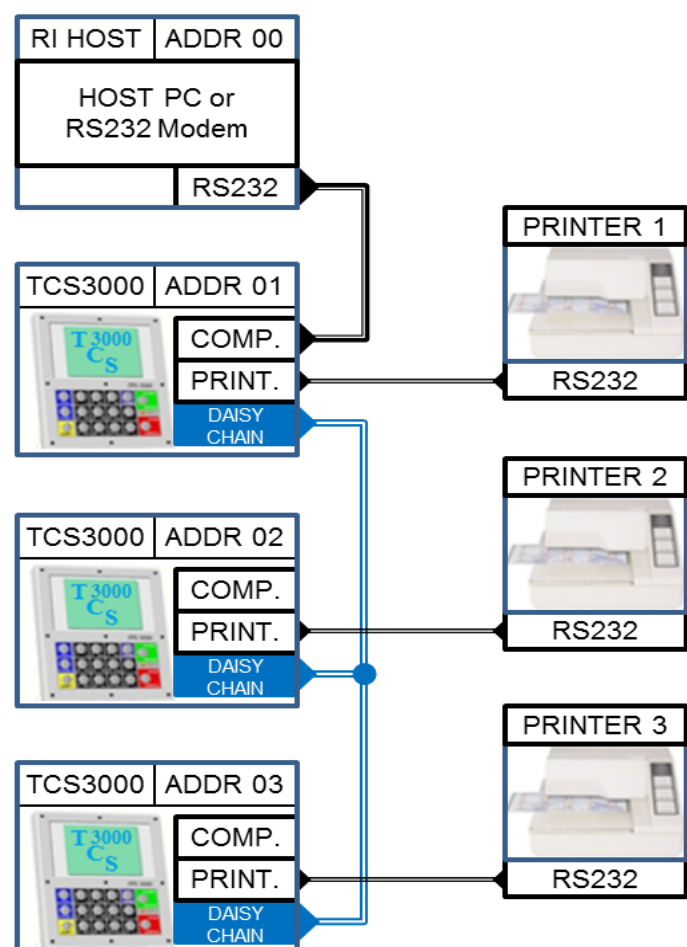
CONFIDENTIAL

EXAMPLE CONNECTION 3: 3 DEVICES CHAIN
WITH RS232 REMOTE CONNECTION AND TCS3000 PRINT HOST



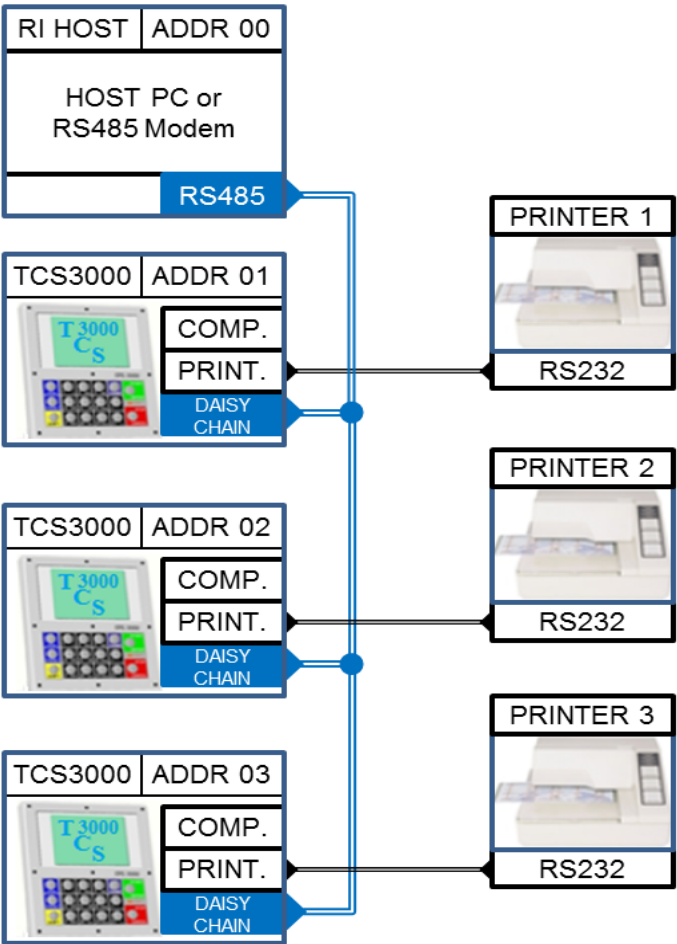
CONFIDENTIAL

EXAMPLE CONNECTION 4: 3 DEVICES CHAIN
WITH RS232 REMOTE CONNECTION AND LOCAL PRINTERS



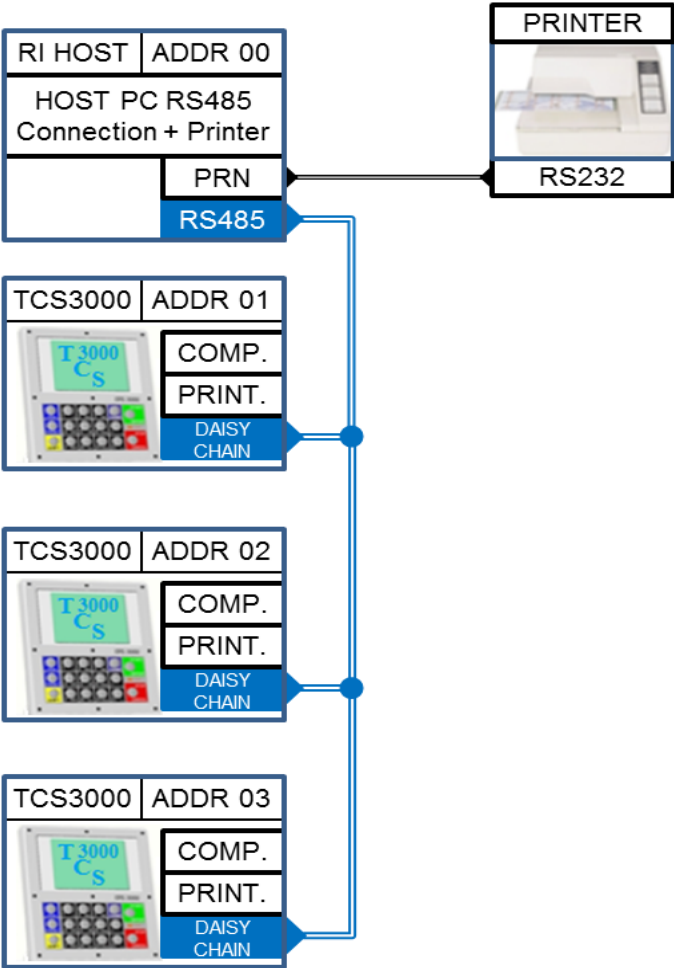
CONFIDENTIAL

EXAMPLE CONNECTION 5: 3 DEVICES CHAIN
WITH RS 485 REMOTE CONNECTION AND LOCAL PRINTERS



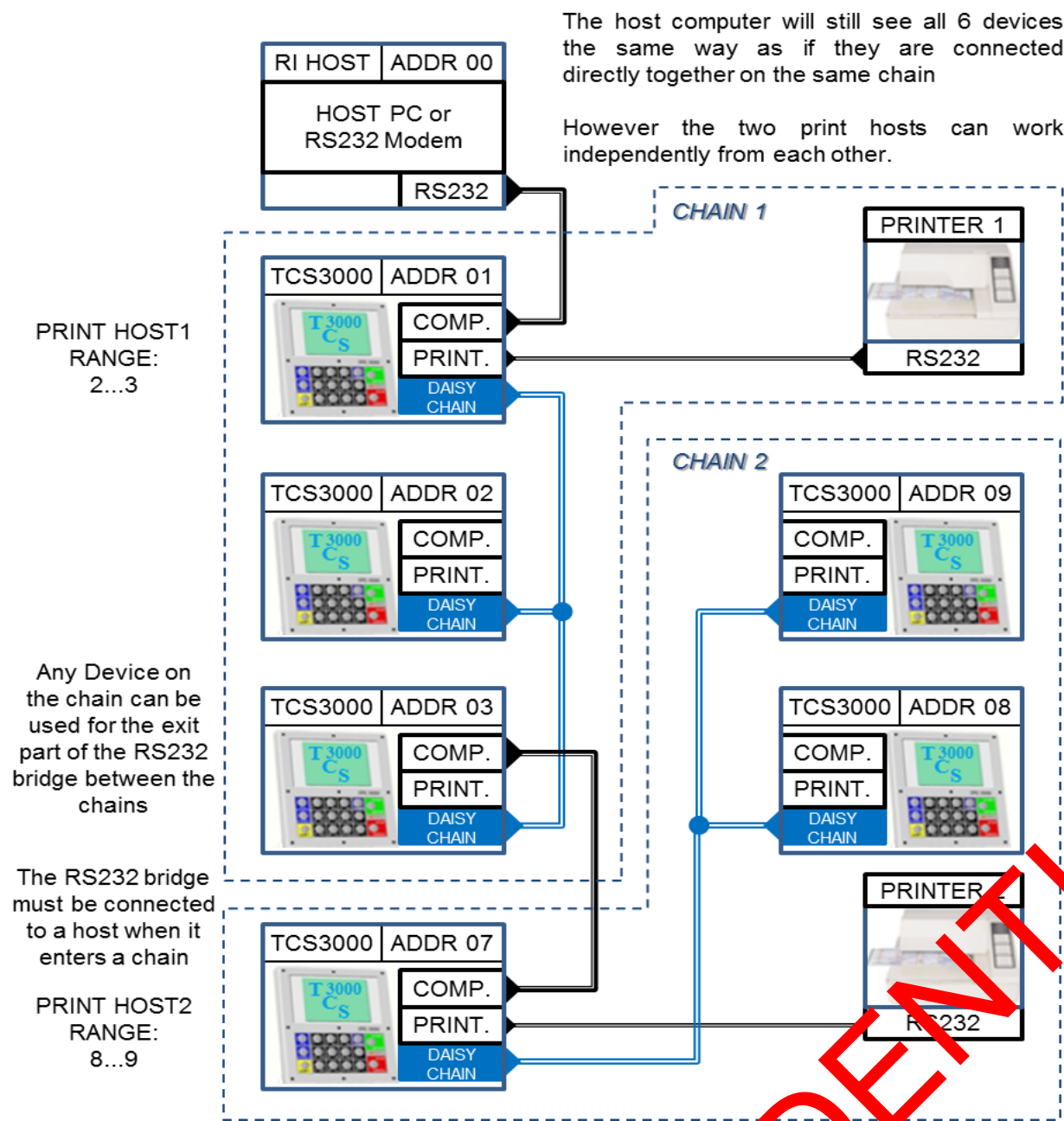
CONFIDENTIAL

EXAMPLE CONNECTION 6: 3 DEVICES CHAIN
WITH RS485 REMOTE CONNECTION AND REMOTE PRINTER



CONFIDENTIAL

**EXAMPLE CONNECTION 7: CLUSTER OF 2CHAINS x 3 DEVICES
WITH REMOTE CONNECTION AND TCS3000 PRINT HOSTS**



CONFIDENTIAL

TCS3000 REMOTE INTERFACE CONCURRENT INTERFACE TIMING

This section applies ONLY to systems using Multiple TCS3000 Devices connected together on the RS485 bus using TCS3000 Print Host AND Remote interface connection through RS232 on the print host.

If you are not using TCS3000 Print host the timing described in this section does not apply to your system. You can use the RS232 remote interface connection without any special restrictions.

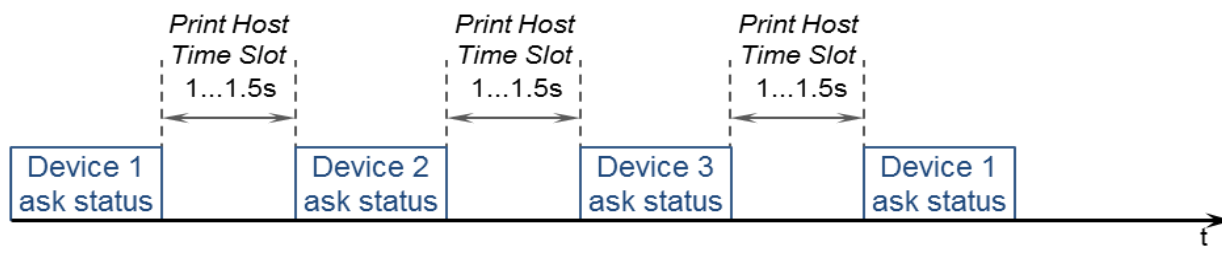
When using a remote connection to chain of two or more TCS3000 devices which are sharing a TCS3000 print host there are some timing guidelines that need to be observed in order to allow the normal operation of both Printer host and remote interface communication:

- 1) When communicating with any device on the chain keep in mind that you are not alone on the RS485 bus – the printer host shares the same bus with you.
- 2) When the chain is “idle” and you are only polling the statuses of the devices allow gaps (Print Host Time slots) every now and then of at least 1 to 1.5 seconds in order for the host to be able to communicate with the devices and eventually print if there is a device print request
- 3) In case you need to perform a data transfer from a device on the chain – perform the whole transfer without leaving any gaps (in order to finish it as fast as possible) and leave a gap after the transfer is completed.
- 4) Due to the operation of the Printer Host and the way the RS232 ↔ RS485 interface bridge operates sometimes even if there are no print requests your data might be delayed with up to 500ms. In order to avoid data errors please use communication timeout of at least 1500ms.
- 5) In case you try to communicate with a device on the chain exactly at the same time when the host is starting a “PRINT” data transfer your data might get delayed more than the allowed internal interface timeout. In this situation you will receive back interface error 252 (0xFC) “BUS BUSY”, just retry sending the command again.
- 6) As a rule of the thumb for idle polling you can use the following scheme:
 - Use communication timeout of at least 1500ms
 - When polling leave gaps of at least 1000 – 1500ms between the polling cycles (or between the calls to each device, which will slow down the polling cycle but will improve the “PRINT” button response time on the TCS3000 devices).
 - When actively transferring data to the devices do not leave print gaps – just finish the transfer as fast as possible and then leave a gap.
 - Include handling of the BUS BUSY error in your code

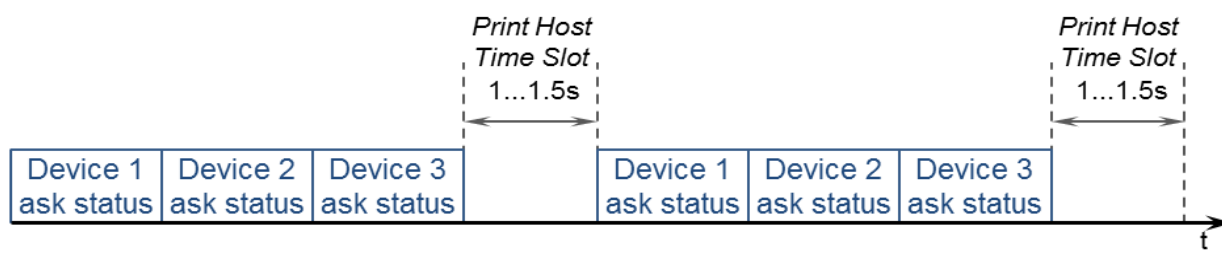
On the next page are shown couple of idle polling timing examples of communication with chain consisting of 3 devices.

TCS3000 REMOTE INTERFACE IDLE POLLING TIMING EXAMPLES

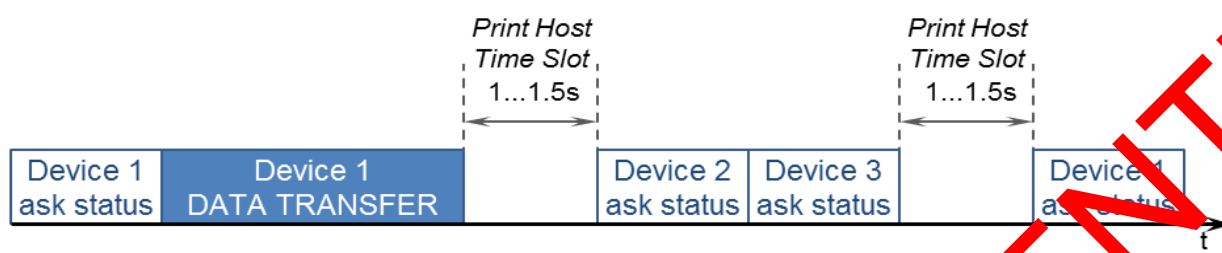
EXAMPLE 1: Idle status polling without data transfer and Print slot after every device



EXAMPLE 2: Idle status polling without data transfer and Print slot after every cycle



EXAMPLE 3: Idle status polling with data transfer and Print slot after every cycle

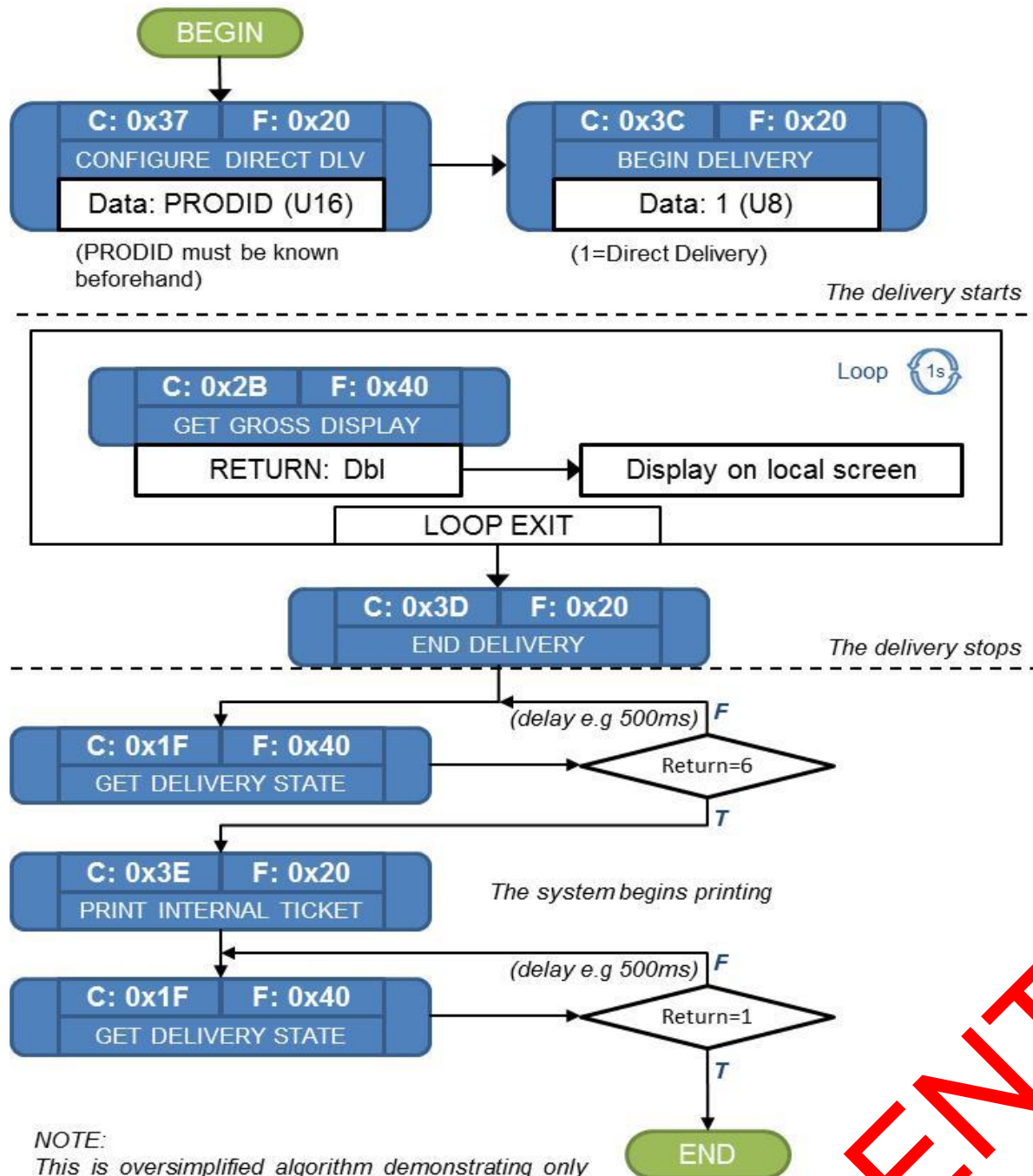


CONFIDENTIAL

TCS3000 REMOTE INTERFACE SOME SIMPLE ALGORITHM EXAMPLES

CONFIDENTIAL

EXAMPLE: SIMPLE DIRECT DELIVERY WITH DEFAULT PRINT

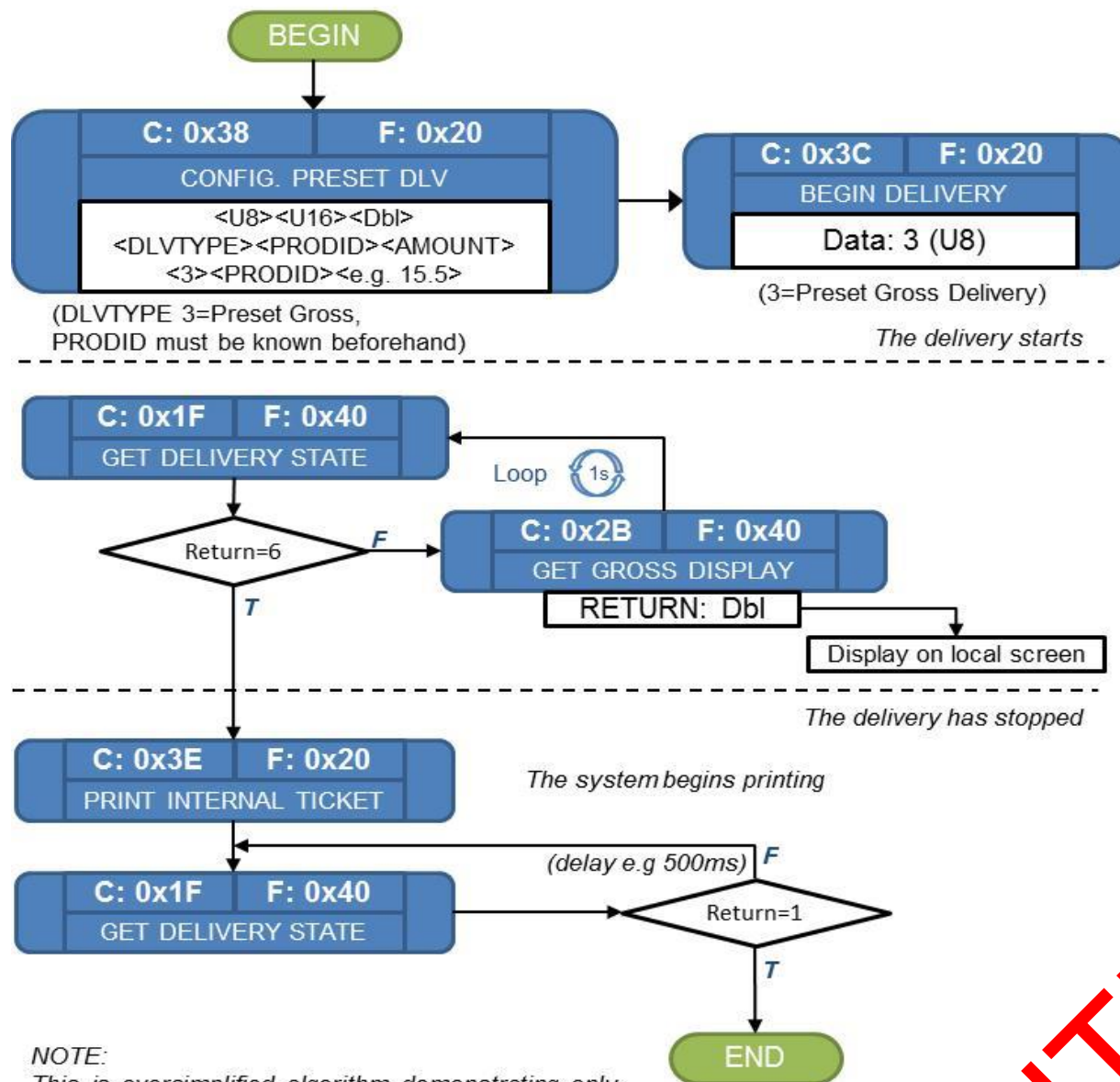


NOTE:

This is oversimplified algorithm demonstrating only the basic concept of the delivery. In practice the programmer will need to add more checks (e.g. Monitoring of the print status through command 0x4F) and conditions for exiting the loops!

CONFIDENTIAL

EXAMPLE: SIMPLE PRESET GROSS DELIVERY WITH DEFAULT PRINT

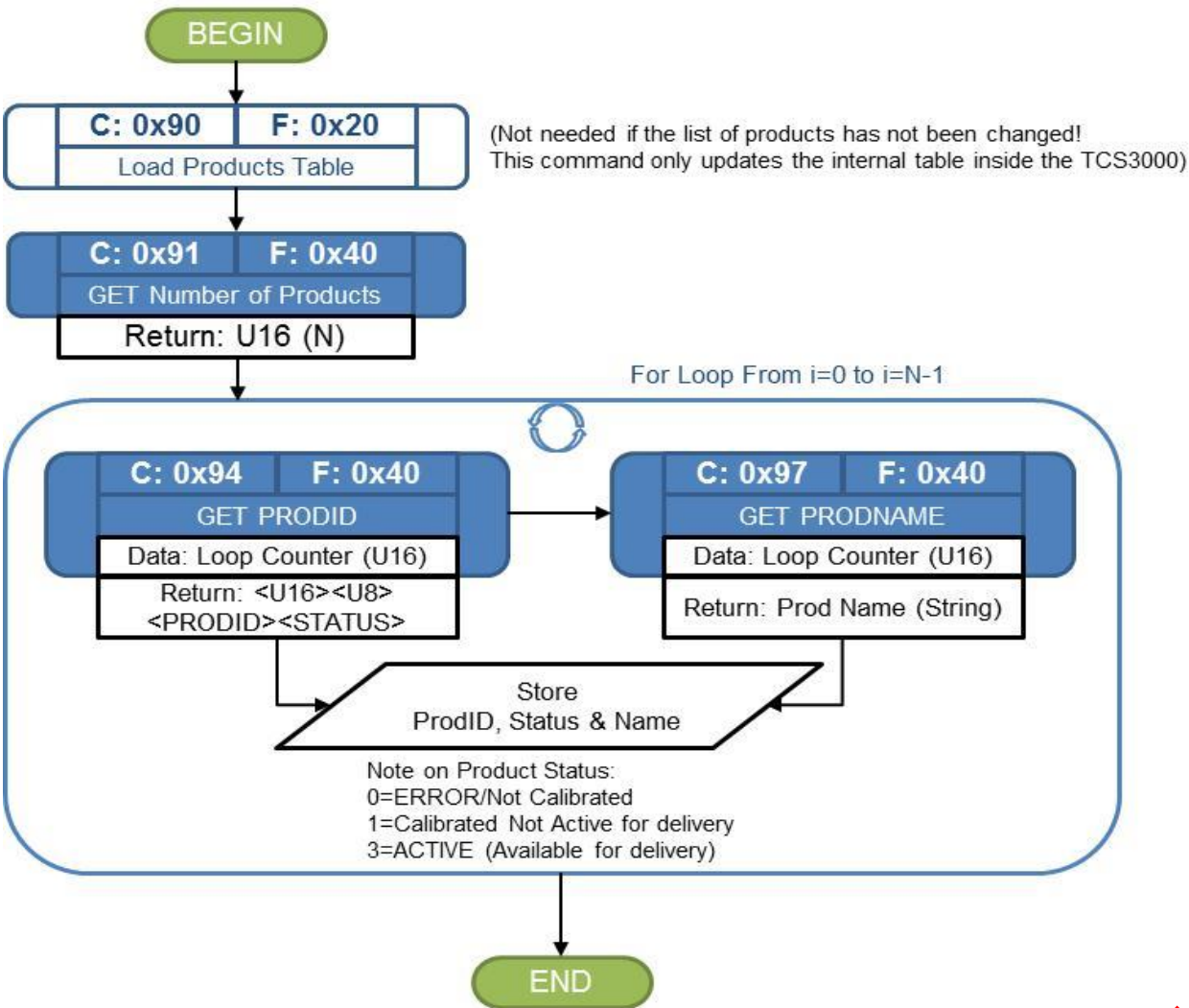


NOTE:

This is oversimplified algorithm demonstrating only the basic concept of the delivery. In practice the programmer will need to add more checks (e.g. Monitoring of the print status through command 0x4F) and conditions for exiting the loops!

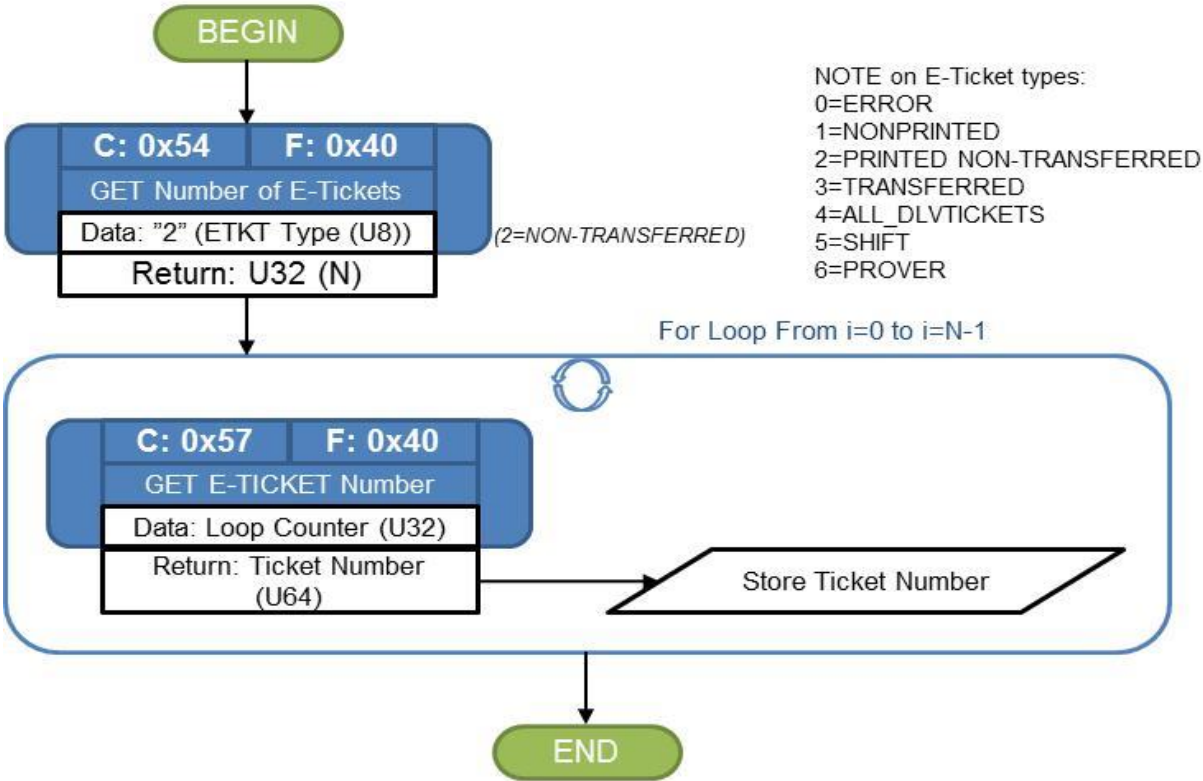
CONFIDENTIAL

EXAMPLE: GET PRODUCTS LIST AND PRODUCT NAMES



CONFIDENTIAL

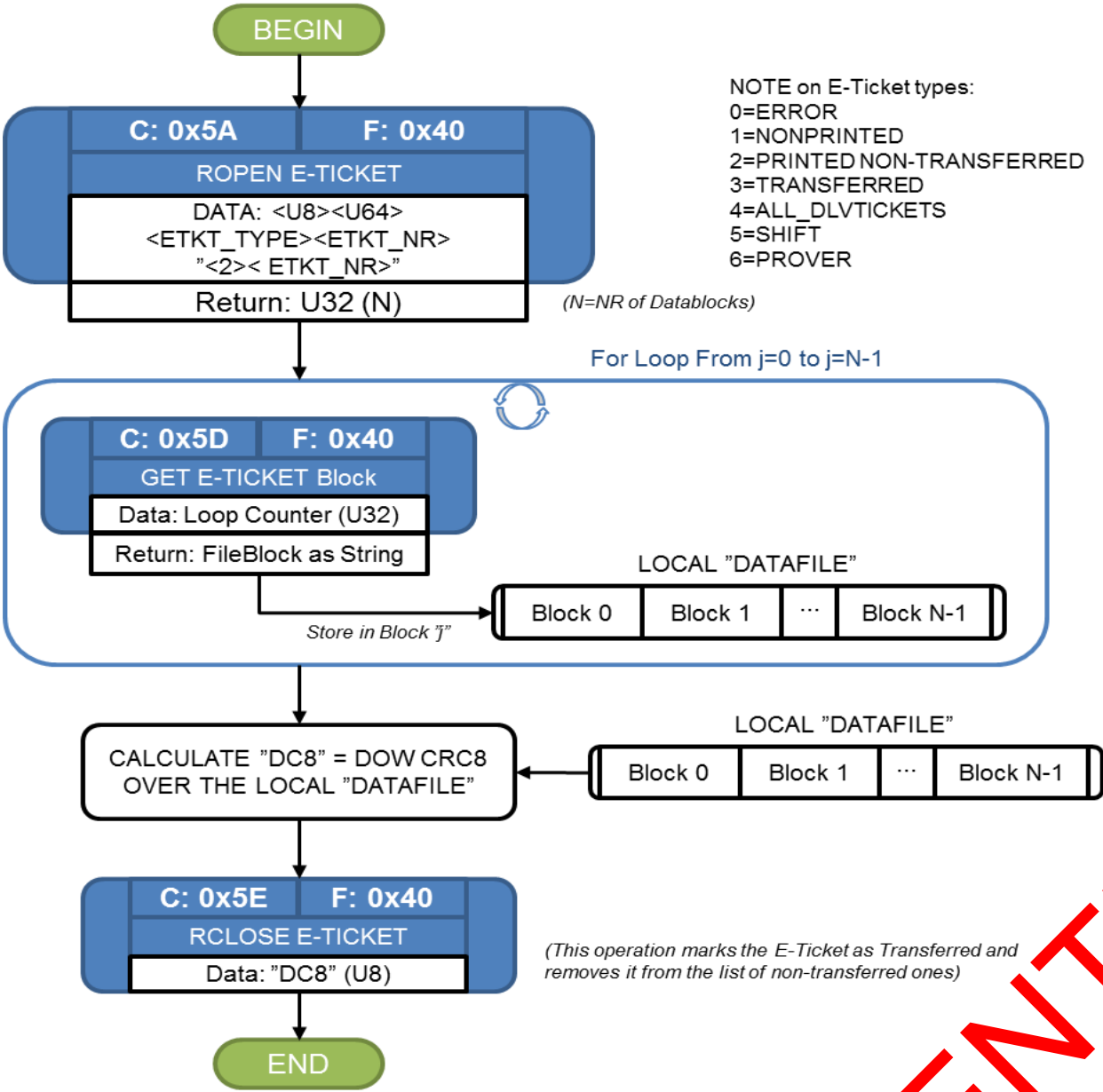
EXAMPLE: GET PRINTED NON-TRANSFERRED E-TICKETS LIST



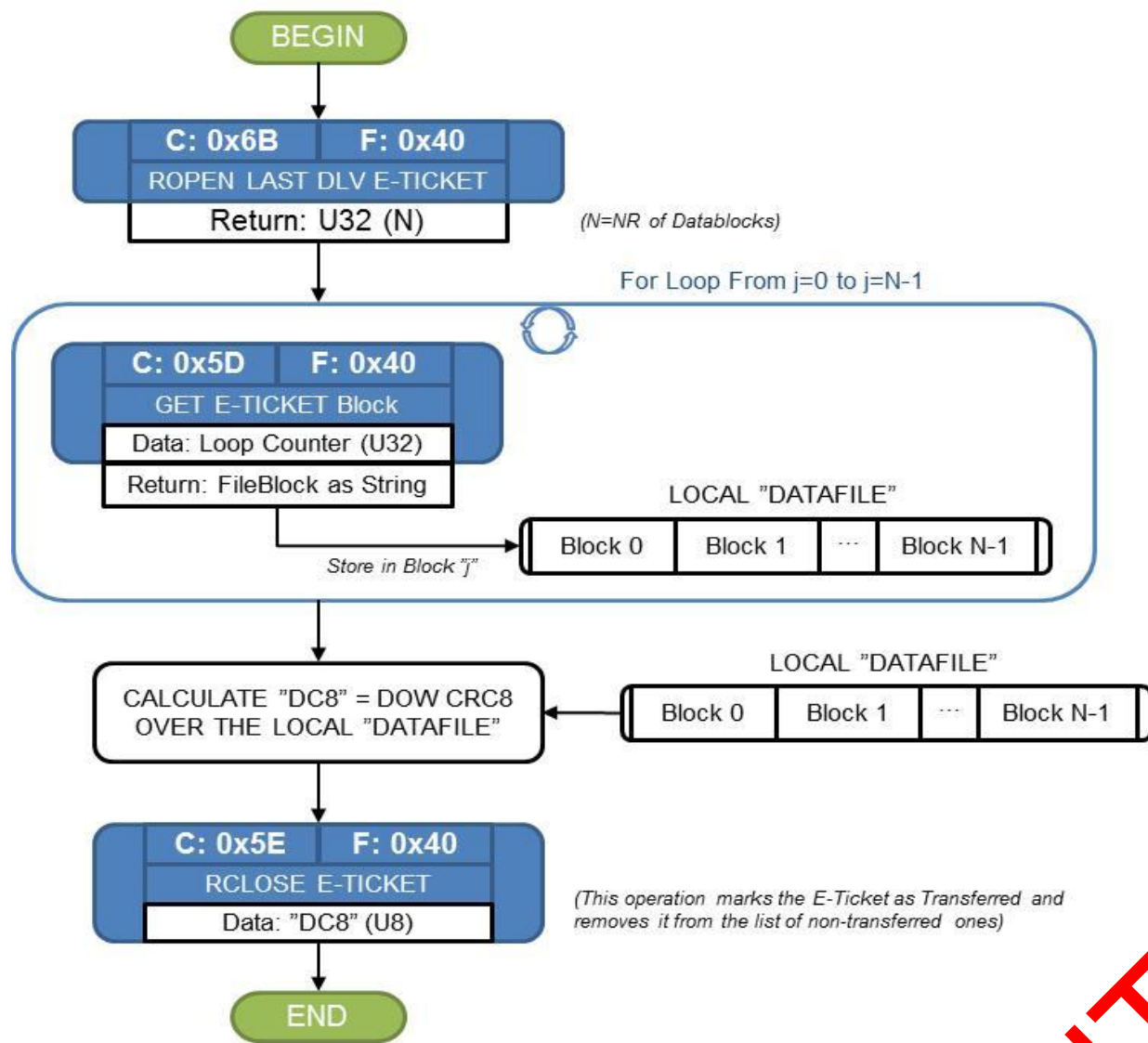
CONFIDENTIAL

EXAMPLE: GET DELIVERY E-TICKET FROM TCS3000

In case only the E-Ticket of the **last** delivery is needed, to avoid asking the system for the full list of tickets and avoid the operations from the previous example use the simpler example on the next page.

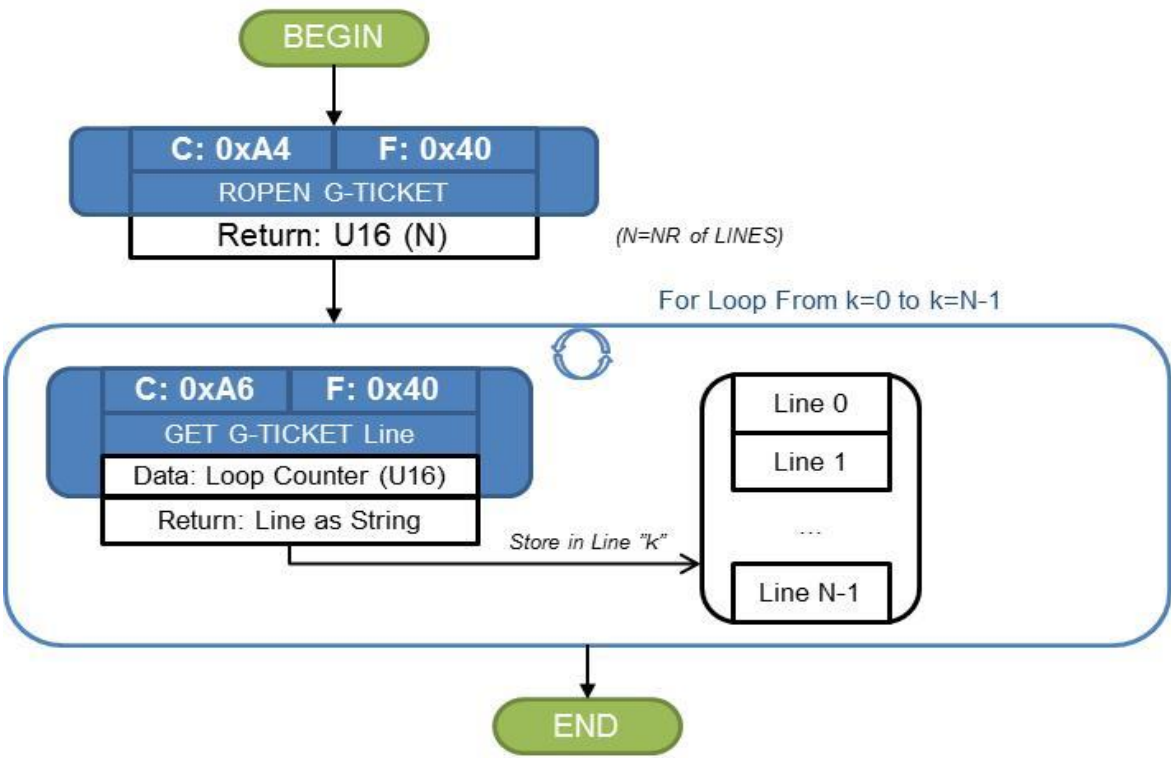


EXAMPLE: GET LATEST DELIVERY E-TICKET FROM TCS3000



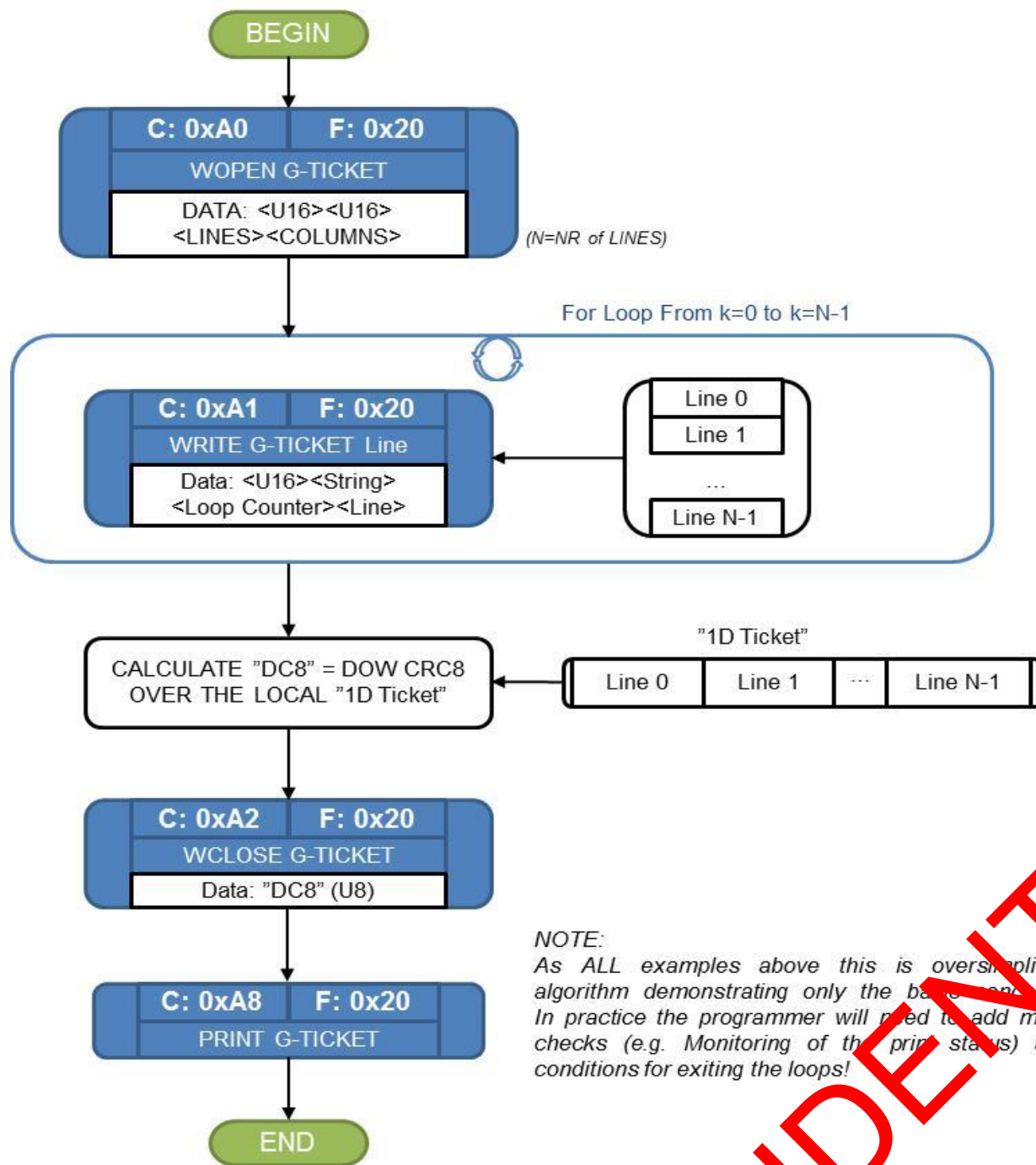
CONFIDENTIAL

EXAMPLE: GET G-TICKET FROM TCS3000



CONFIDENTIAL

EXAMPLE: WRITE AND PRINT G-TICKET IN TCS3000



CONFIDENTIAL